



US012401393B2

(12) **United States Patent**
Poulton et al.

(10) **Patent No.:** **US 12,401,393 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CONFIGURABLE, POWER SUPPLY
VOLTAGE REFERENCED SINGLE-ENDED
SIGNALING WITH ESD PROTECTION**

(71) Applicant: **Rambus Inc.**, San Jose, CA (US)

(72) Inventors: **John W. Poulton**, Chapel Hill, NC
(US); **Frederick A. Ware**, Los Altos
Hills, CA (US); **Carl W. Werner**, Los
Gatos, CA (US)

(73) Assignee: **Rambus Inc.**, San Jose, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 134 days.

(21) Appl. No.: **18/195,524**

(22) Filed: **May 10, 2023**

(65) **Prior Publication Data**

US 2023/0412213 A1 Dec. 21, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/354,235, filed on
Jun. 22, 2021, now Pat. No. 11,689,246, which is a
(Continued)

(51) **Int. Cl.**
H04B 3/56 (2006.01)
G06F 13/40 (2006.01)
H03F 3/24 (2006.01)
H04B 3/54 (2006.01)
H04L 25/02 (2006.01)
H04B 1/04 (2006.01)
H04B 10/73 (2013.01)

(Continued)

(52) **U.S. Cl.**
CPC **H04B 3/56** (2013.01); **G06F 13/4072**
(2013.01); **H03F 3/24** (2013.01); **H04B 3/54**
(2013.01); **H04L 25/0272** (2013.01); **H01L**

2224/48091 (2013.01); **H01L 2224/48227**
(2013.01); **H01L 2224/49109** (2013.01); **H01L**
2924/13091 (2013.01); **H01L 2924/15311**
(2013.01); **H04B 2001/0408** (2013.01); **H04B**
10/0731 (2013.01); **H04B 10/40** (2013.01);
H04B 10/50 (2013.01)

(58) **Field of Classification Search**

CPC **H04B 3/56**; **H04B 3/54**; **H04B 2001/0408**;
H04B 10/50; **H04B 10/0731**; **H04B**
10/40; **H03F 3/24**; **H04L 25/0272**; **G06F**
13/4072

USPC **361/56**, **91.1**, **111**
See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,644,255 A 7/1997 Taylor
5,684,429 A 11/1997 Sanwo et al.
(Continued)

OTHER PUBLICATIONS

EP Examination Report dated Feb. 25, 2014 in EP Application No.
10728464.8. 5 pages.

(Continued)

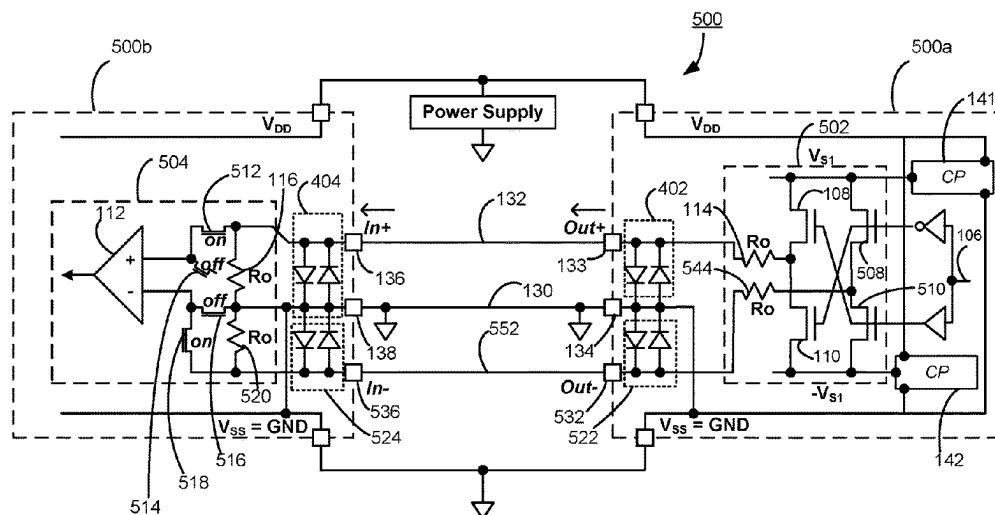
Primary Examiner — Danny Nguyen

(74) *Attorney, Agent, or Firm* — The Neudeck Law Firm,
LLC

(57) **ABSTRACT**

A single-ended data transmission system transmits a signal
having a signal voltage that is referenced to a power supply
voltage and that swings above and below the power supply
voltage. The power supply voltage is coupled to a power
supply rail that also serves as a signal return path. The signal
voltage is derived from two signal supply voltages generated
by a pair of charge pumps that draw substantially same
amount of current from a power supply.

20 Claims, 15 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/685,861, filed on Nov. 15, 2019, now Pat. No. 11,075,671, which is a continuation of application No. 15/890,341, filed on Feb. 6, 2018, now Pat. No. 10,516,442, which is a continuation of application No. 15/138,077, filed on Apr. 25, 2016, now Pat. No. 9,923,602, which is a continuation of application No. 14/072,307, filed on Nov. 5, 2013, now Pat. No. 9,350,421, which is a continuation of application No. 13/318,475, filed as application No. PCT/US2010/029101 on Mar. 29, 2010, now Pat. No. 8,605,397.

- (60) Provisional application No. 61/176,411, filed on May 7, 2009.

(51) **Int. Cl.**

H04B 10/40 (2013.01)

H04B 10/50 (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,892,717	A	4/1999	Malarsie	
6,005,414	A	12/1999	Reay	
6,184,717	B1	2/2001	Crick	
6,229,845	B1	5/2001	Chan	
6,294,933	B1 *	9/2001	Chun	 H03K 19/0013 326/86
6,320,494	B1	11/2001	Bartels et al.	
6,972,607	B1	12/2005	Chen et al.	
7,009,827	B1	3/2006	Lee et al.	
7,459,938	B2	12/2008	To et al.	
8,605,397	B2	12/2013	Poulton et al.	

2004/0057525	A1	3/2004	Rajan et al.
2004/0238194	A1	12/2004	Barr et al.
2005/0212586	A1	9/2005	Daga et al.
2005/0285620	A1	12/2005	Wehage et al.
2006/0015275	A1	1/2006	Batra et al.
2006/0126745	A1	6/2006	Hag et al.
2007/0053227	A1	3/2007	Ragone et al.
2007/0097658	A1	5/2007	Yang et al.
2008/0048730	A1	2/2008	Bae
2008/0218292	A1	9/2008	Park et al.
2008/0266019	A1	10/2008	Fusayasu et al.
2008/0290976	A1	11/2008	Feldtkeller
2009/0039985	A1	2/2009	Sumita
2009/0219179	A1	9/2009	Bae
2010/0001789	A1	1/2010	Hollis
2011/0193620	A1	8/2011	Hollis

OTHER PUBLICATIONS

EP Invitation to Respond re EP Application No. 10728464.8 dated Dec. 20, 2011. 2 pages.

EP Response dated Apr. 25, 2012 to the Invitation to Respond re EP Application No. 10728464.8 dated Dec. 20, 2011. 8 pages.

Honda, Jun & Adams, Jonathan "Class D Audio Amplifier Basics", Application Note AN-1071, International IOR Rectifier, Application Note dated Feb. 8, 2005. 14 Pages.

Notification of Transmittal of the International Search Report and Written Opinion of the International Searching Authority issued for PCT/US2010/029101, mail date Sep. 6, 2010. 12 pages.

Poulton, John W. et al., "A 1.17-pJ/b, 25-GB/s/pin Ground-Referenced Single-Ended Serial Link for Off- and On-Package Communication Using a Process- and Temperature-Adaptive Voltage Regulator", IEEE Journal of Solid-State Circuits, vol. 54, No. 1, Jan. 2019. pp. 43-54. 12 pages.

* cited by examiner

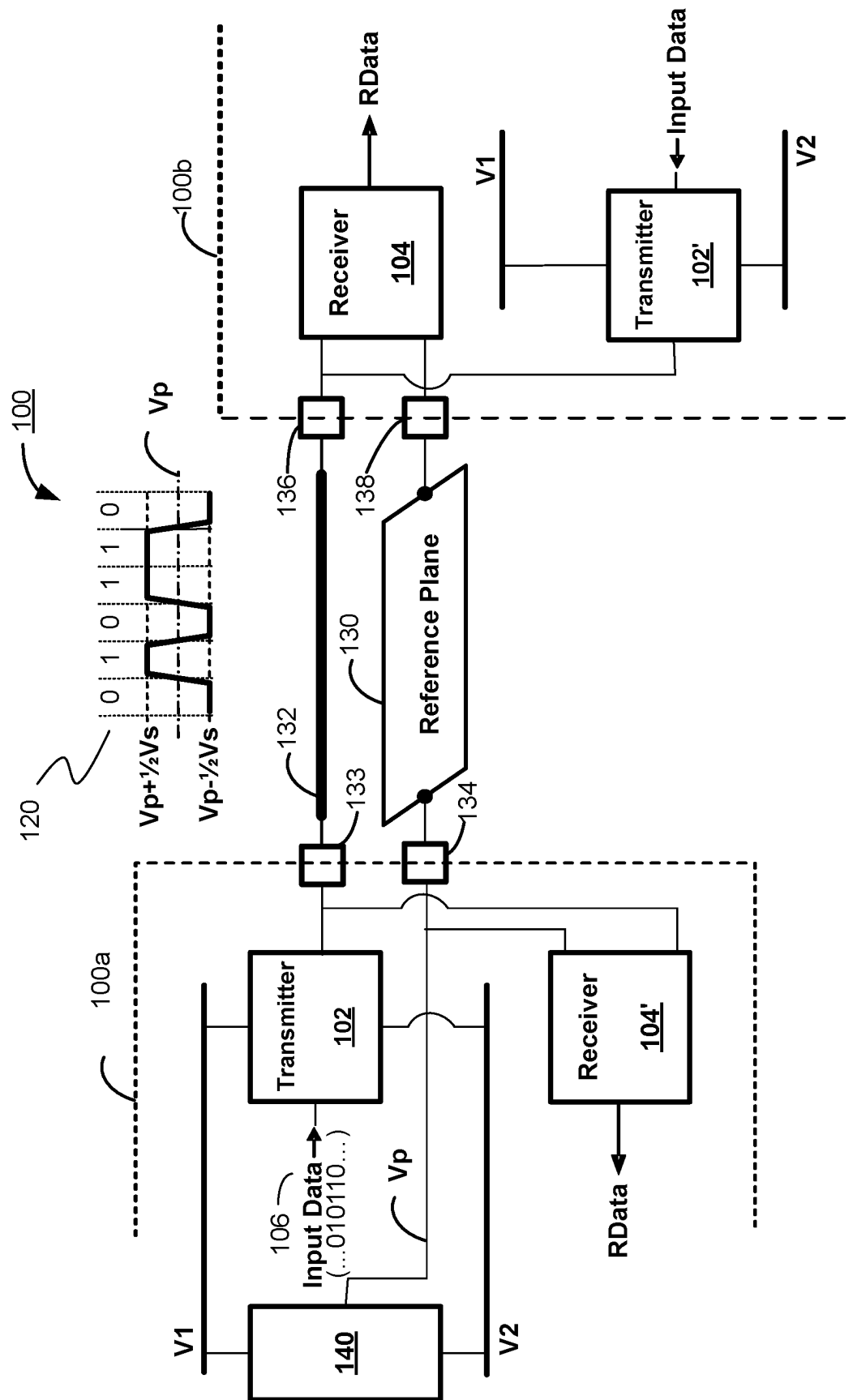


FIG. 1A

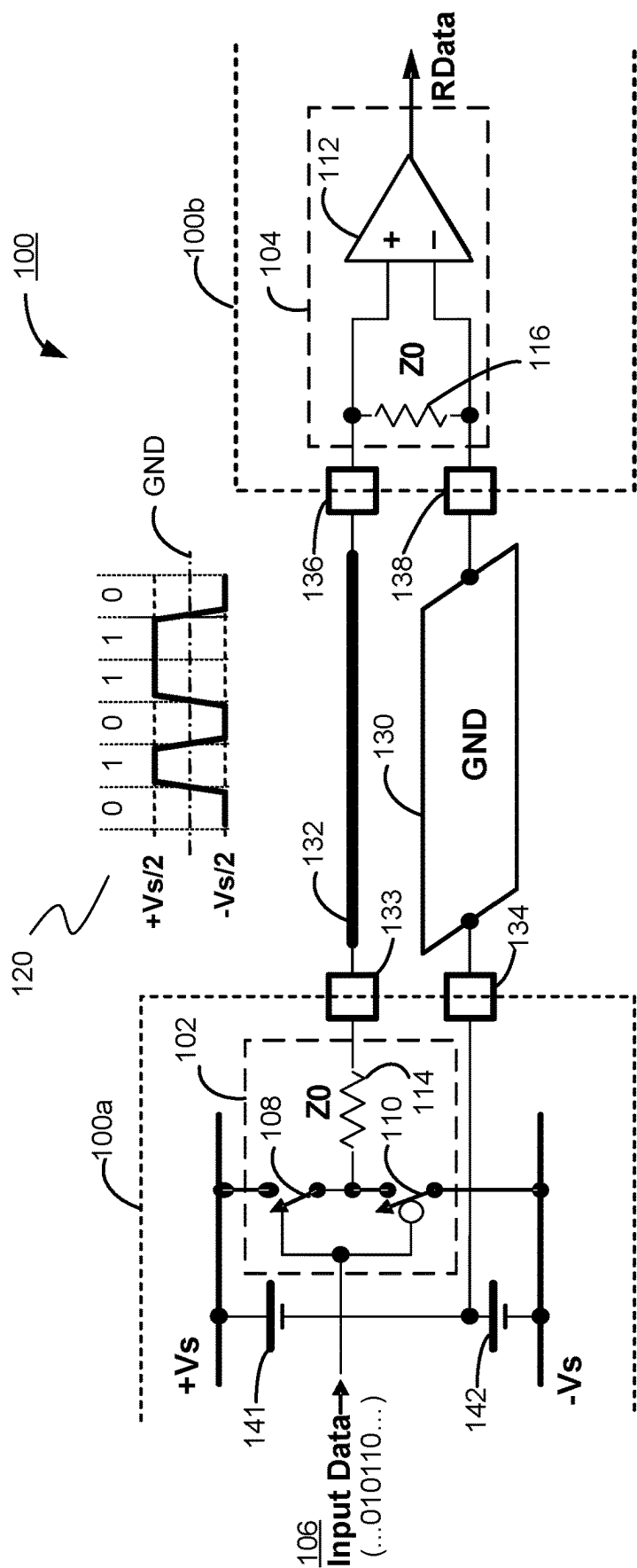


FIG. 1B

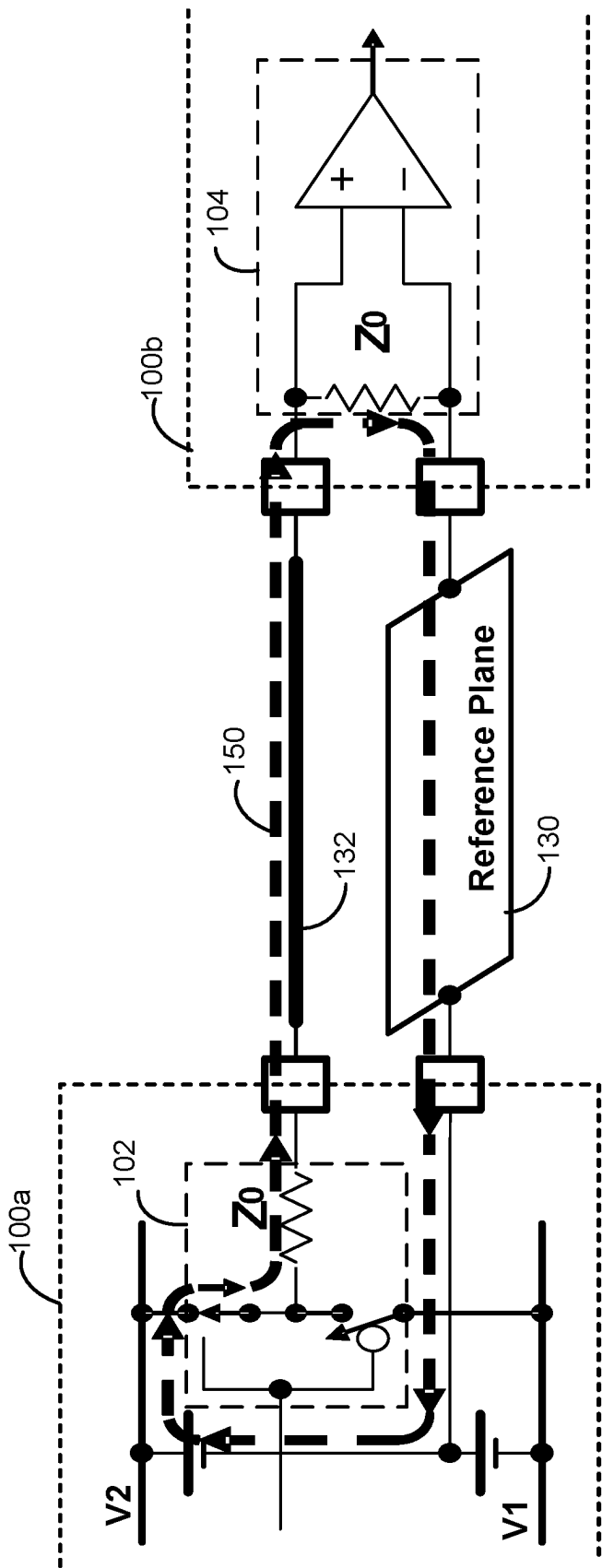


FIG. 1C

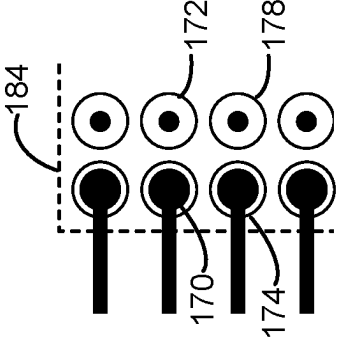
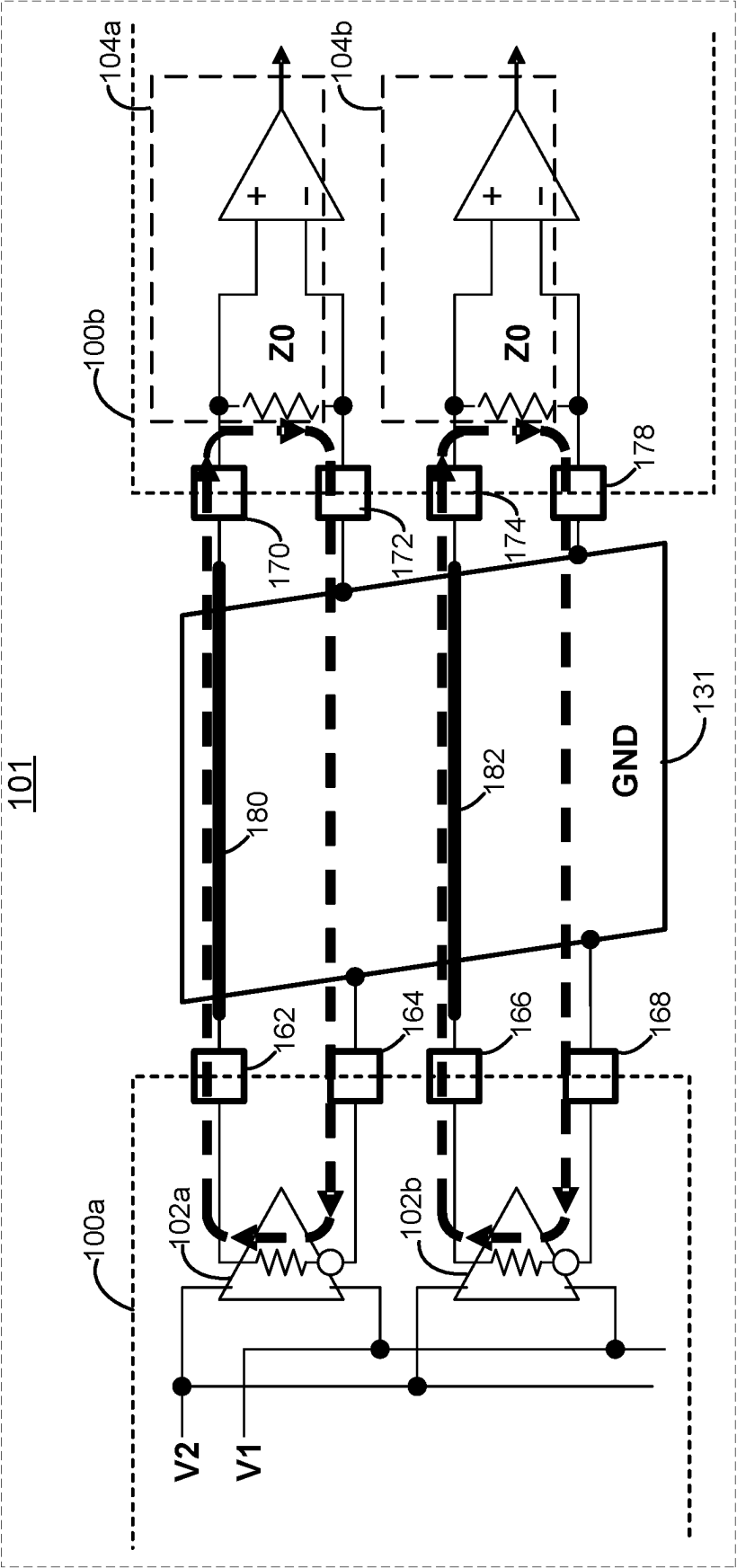


FIG. 1D

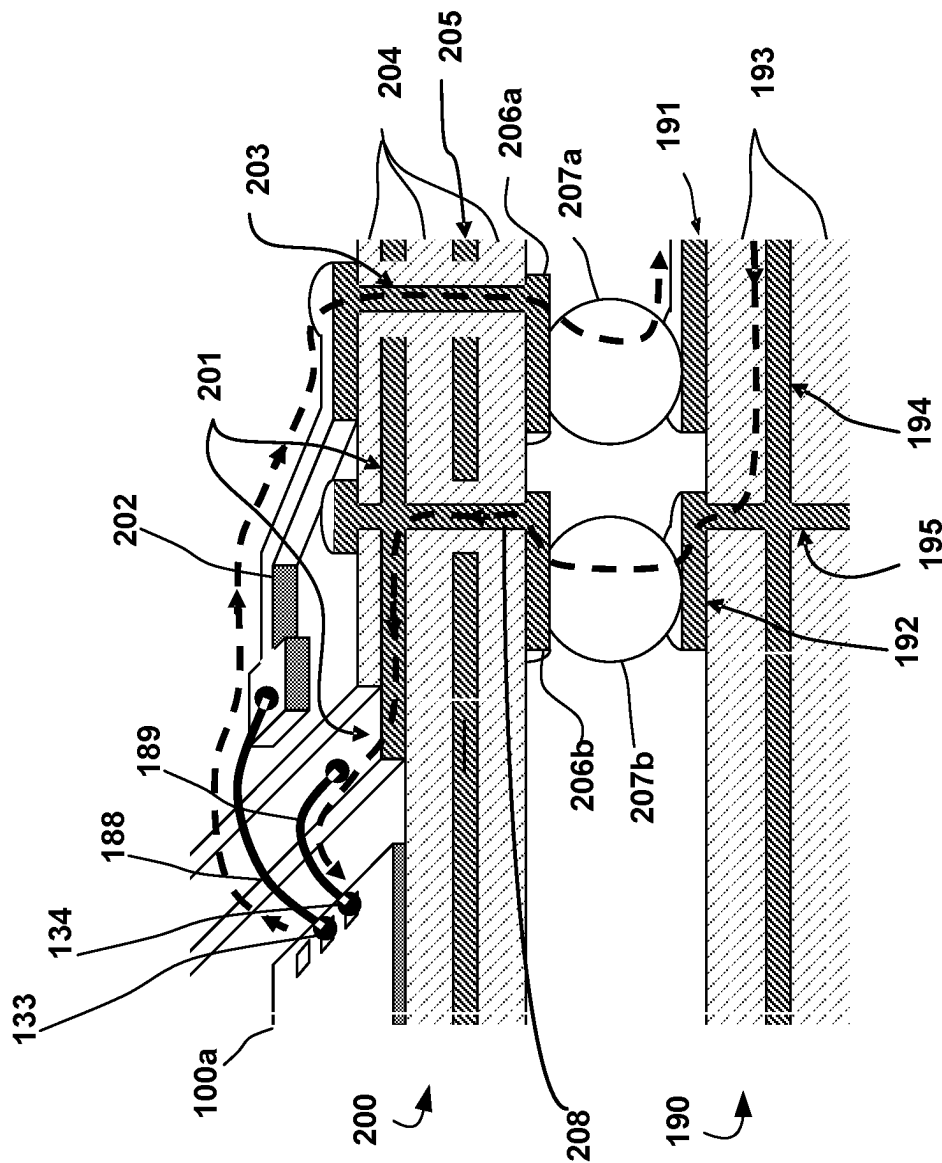


FIG. 1E

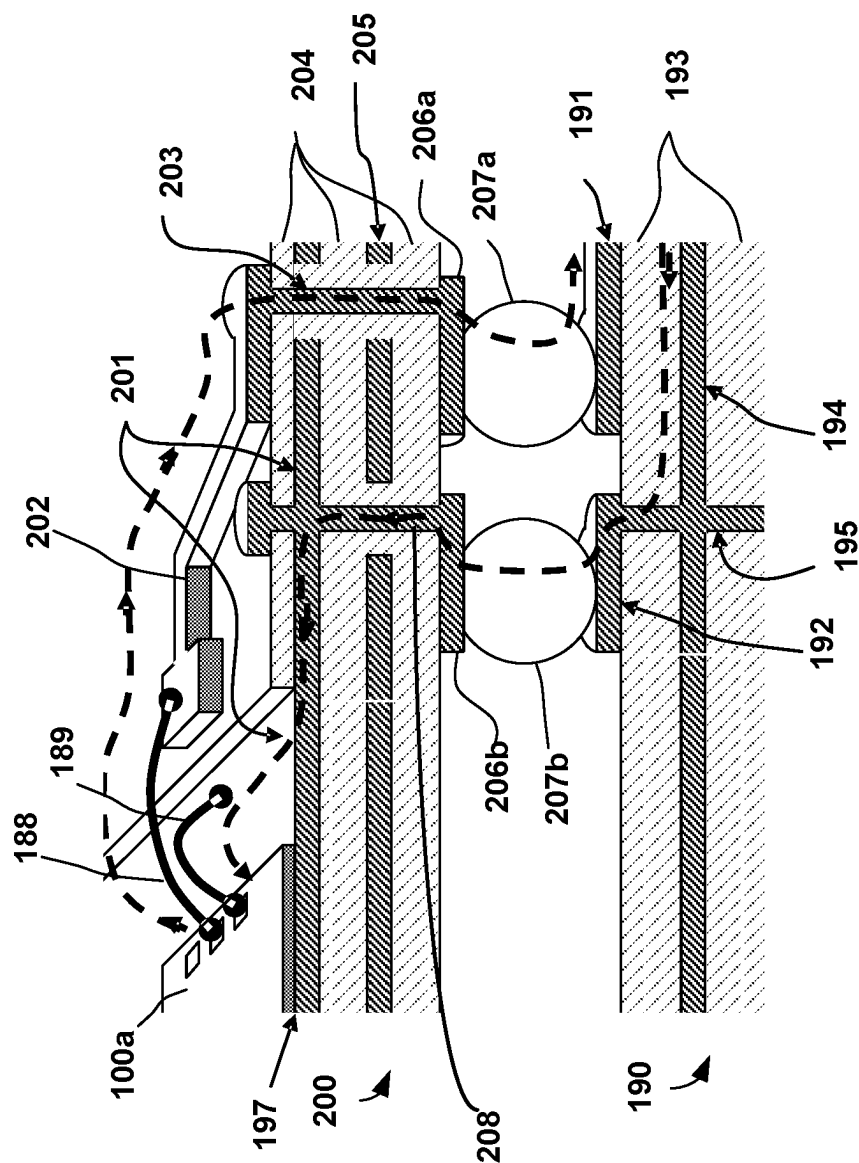


FIG. 1F

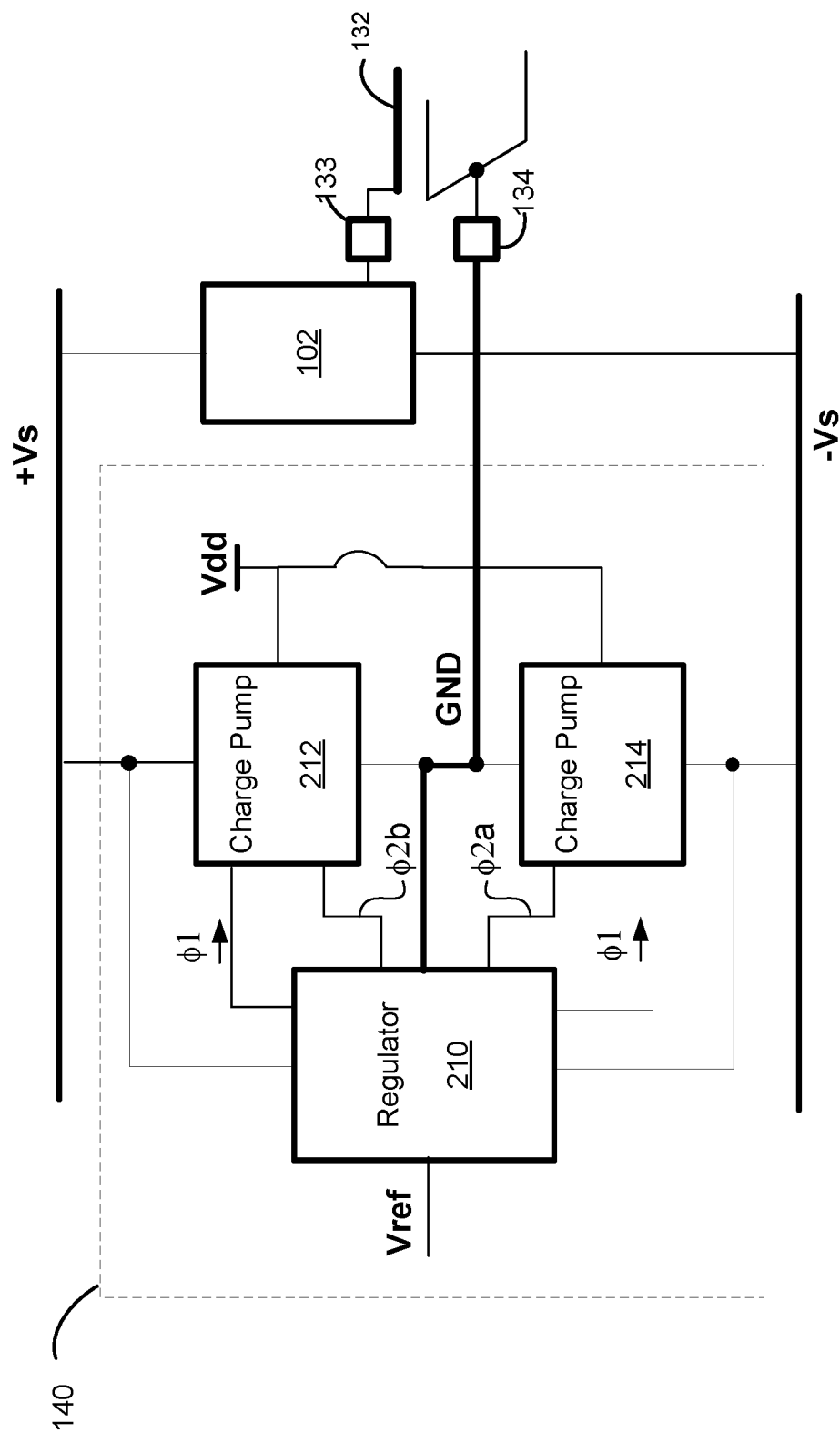


FIG. 2A

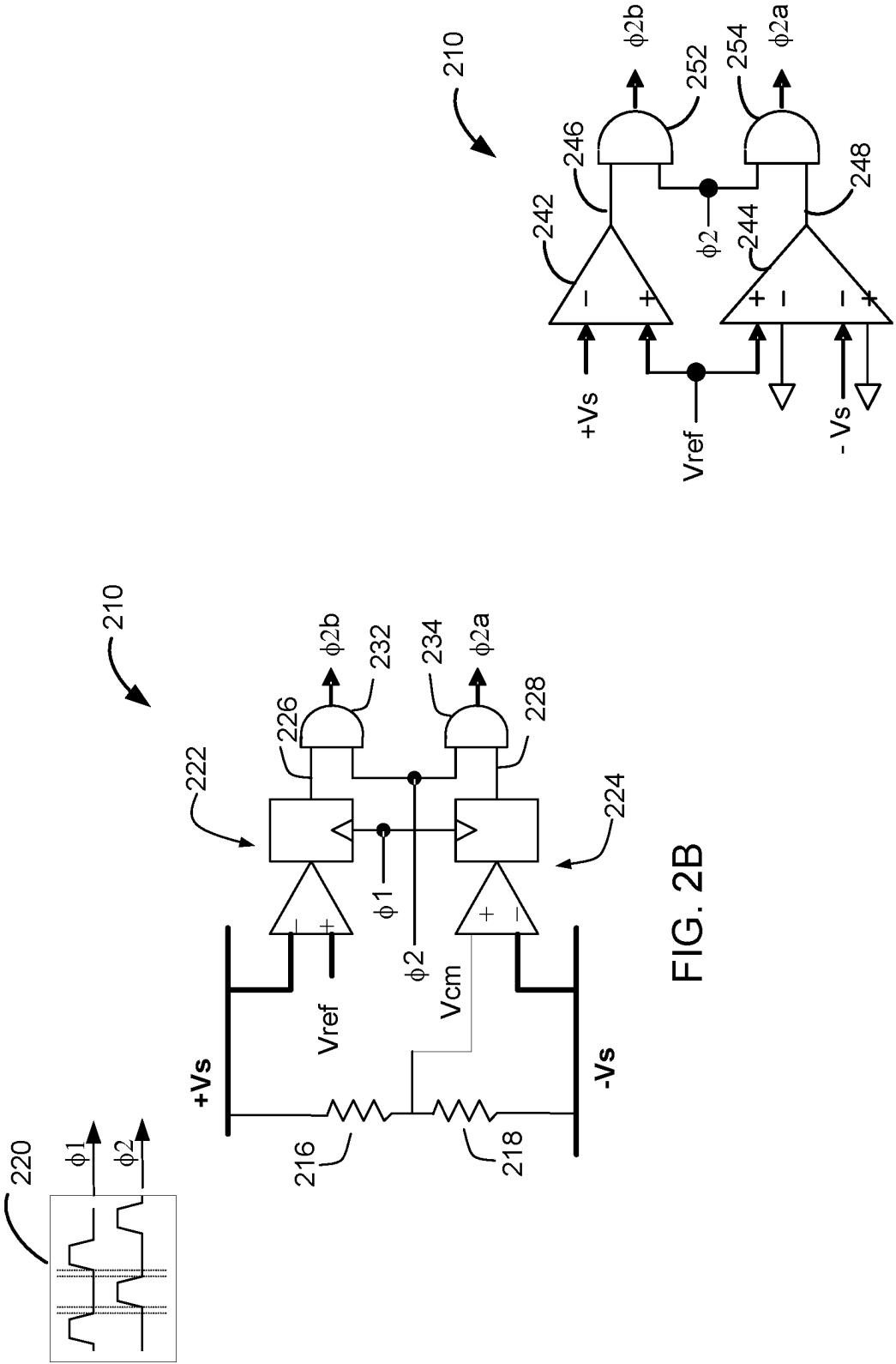


FIG. 2C

FIG. 2B

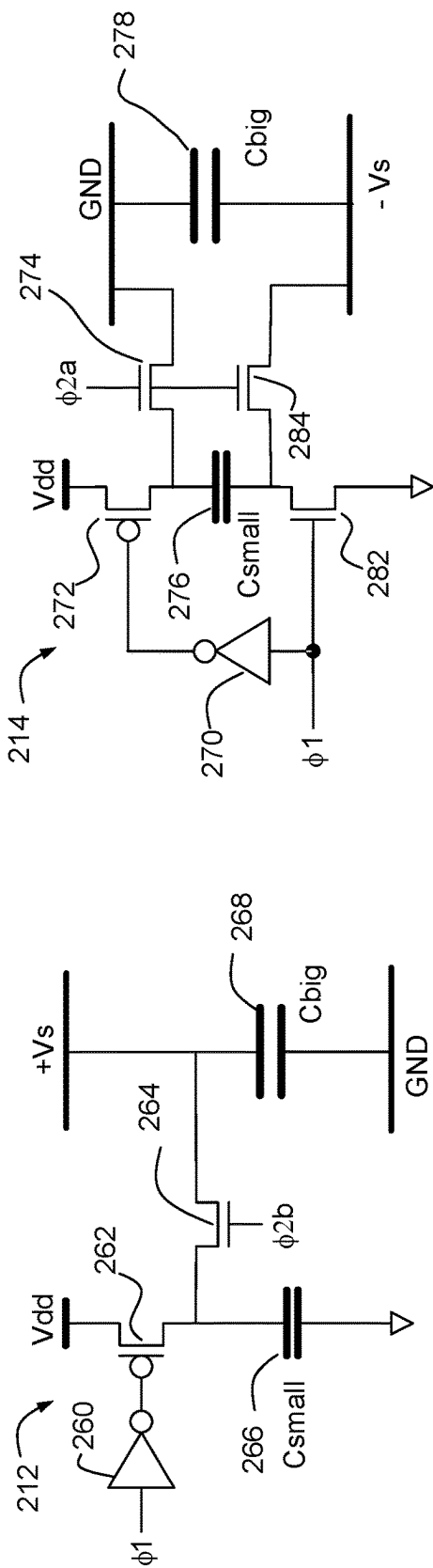


FIG. 2E

FIG. 2D

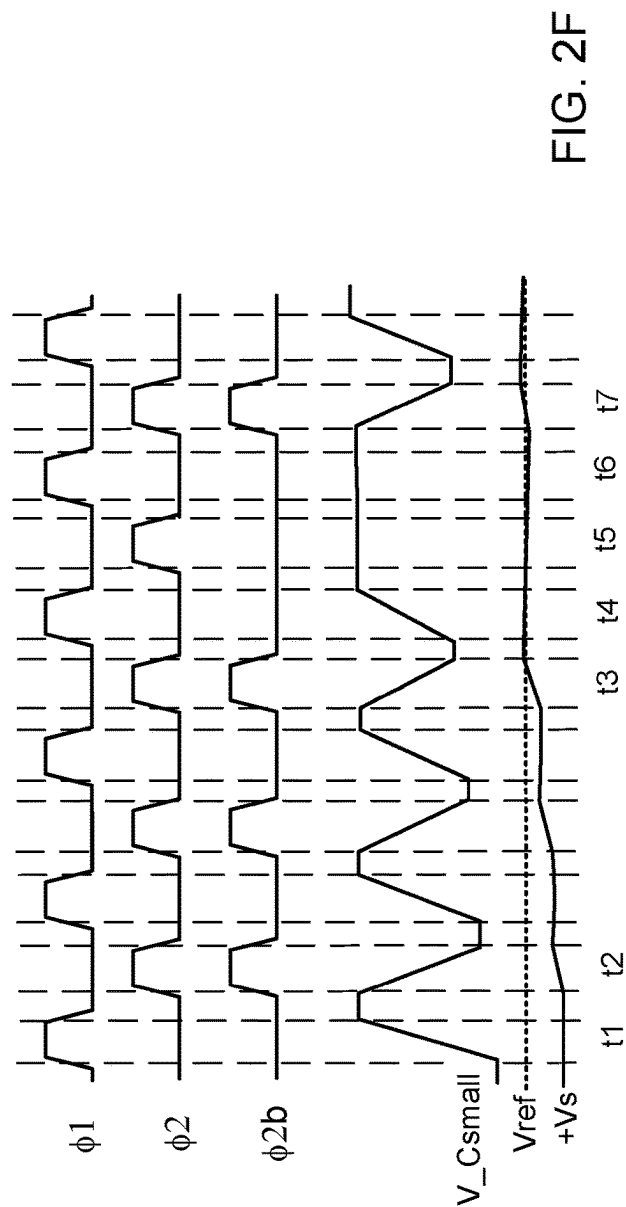


FIG. 2F

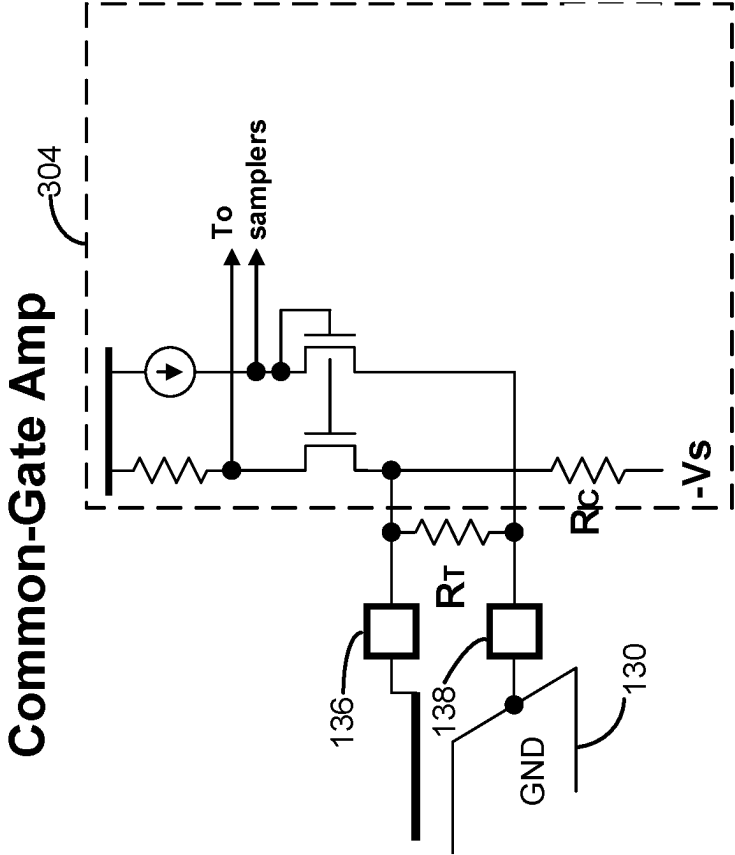


FIG. 3B

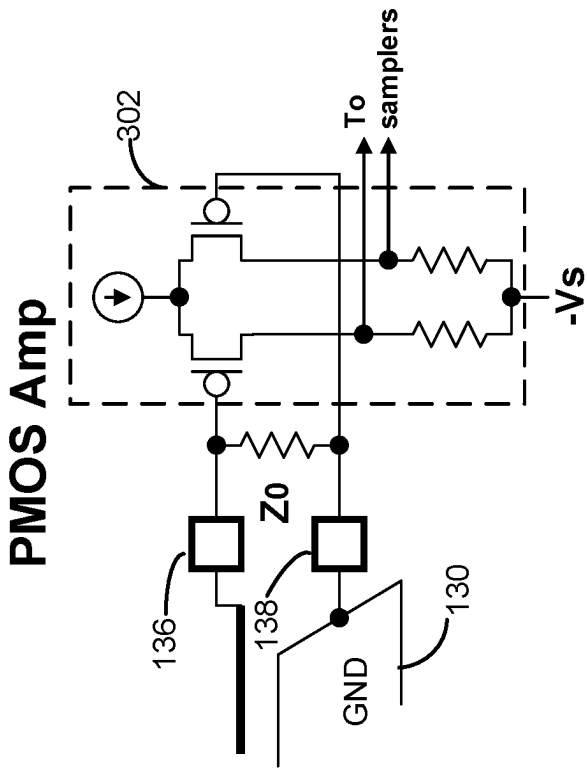


FIG. 3A

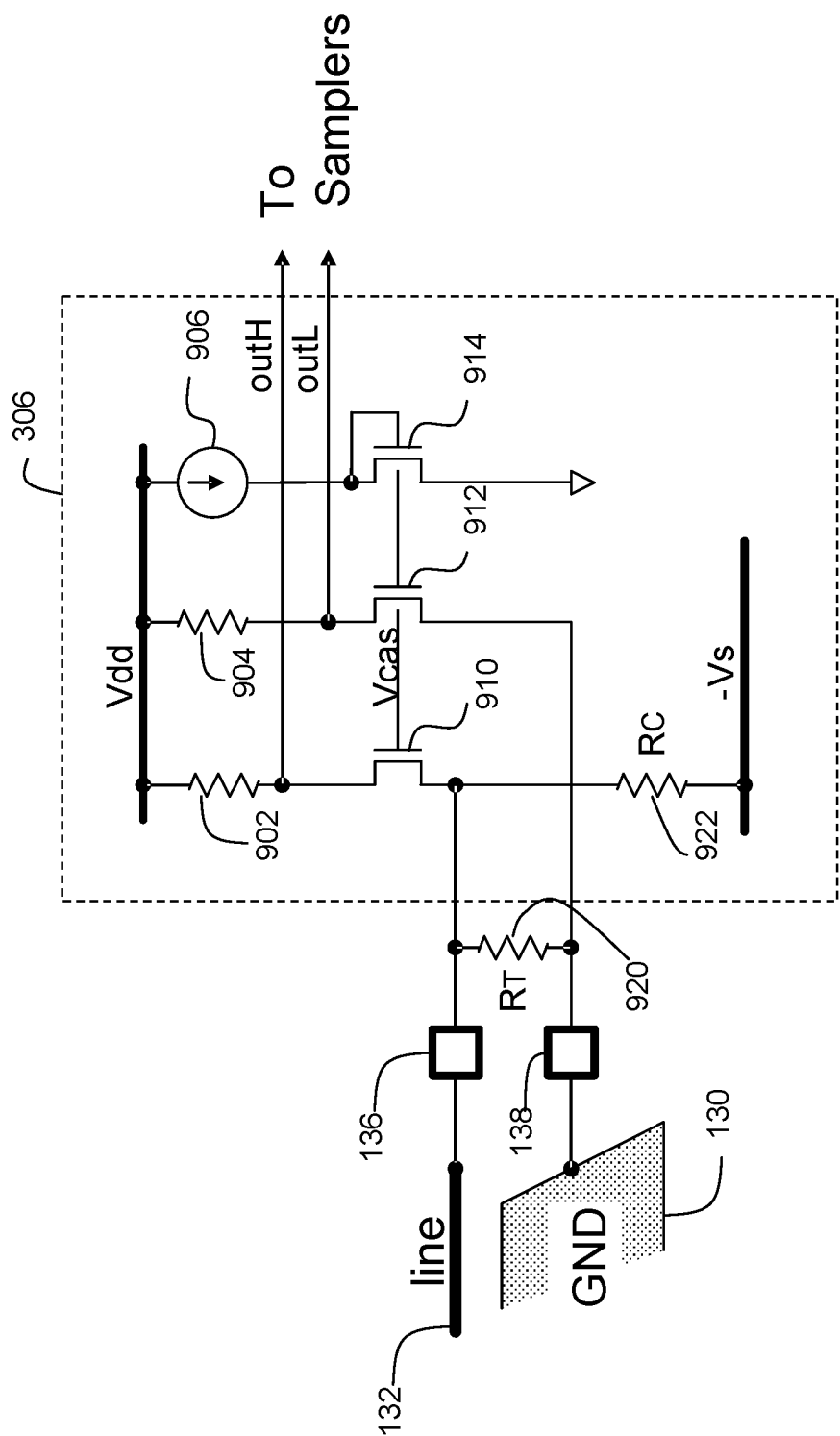


FIG. 3C

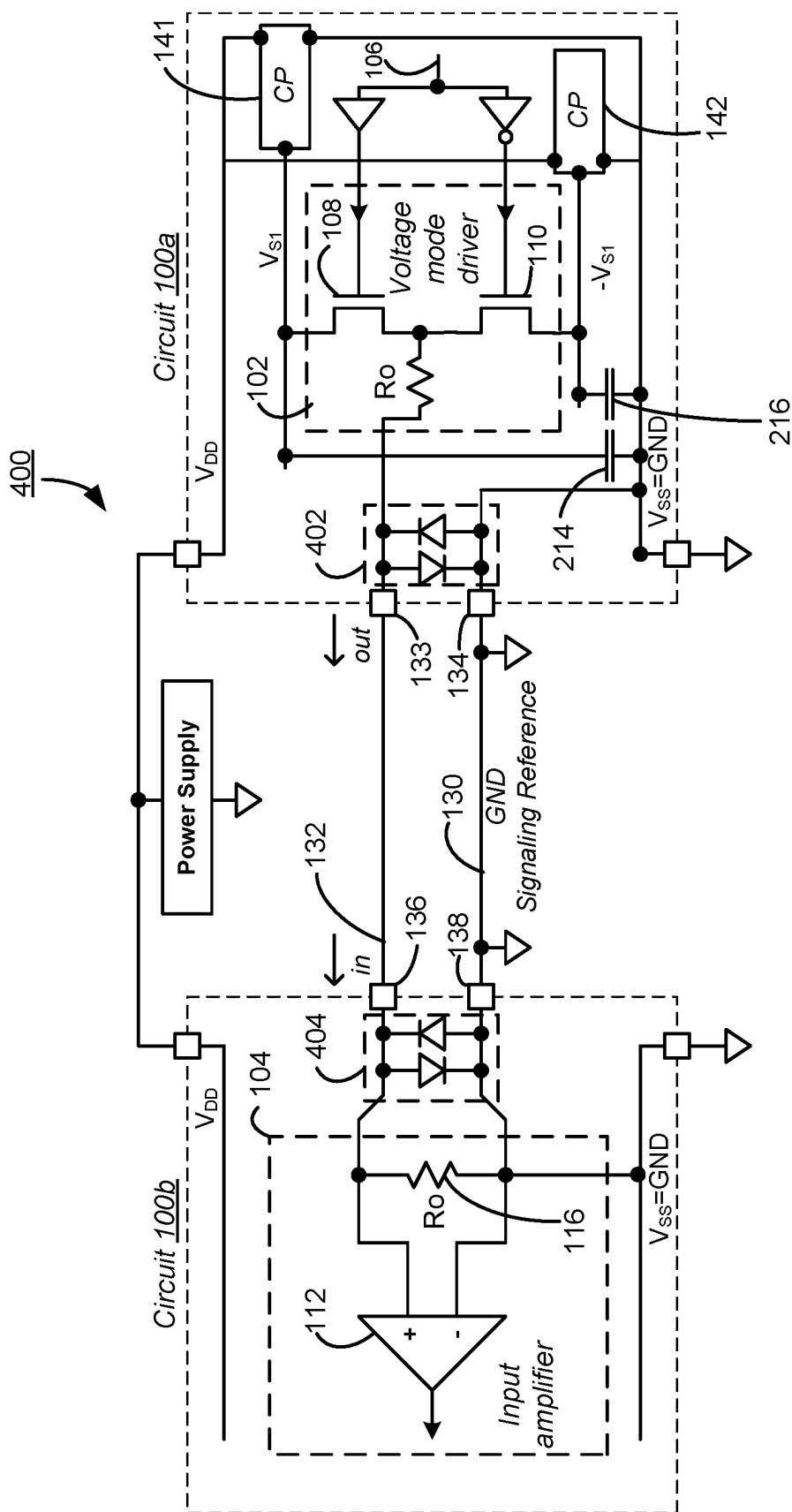


FIG. 4A

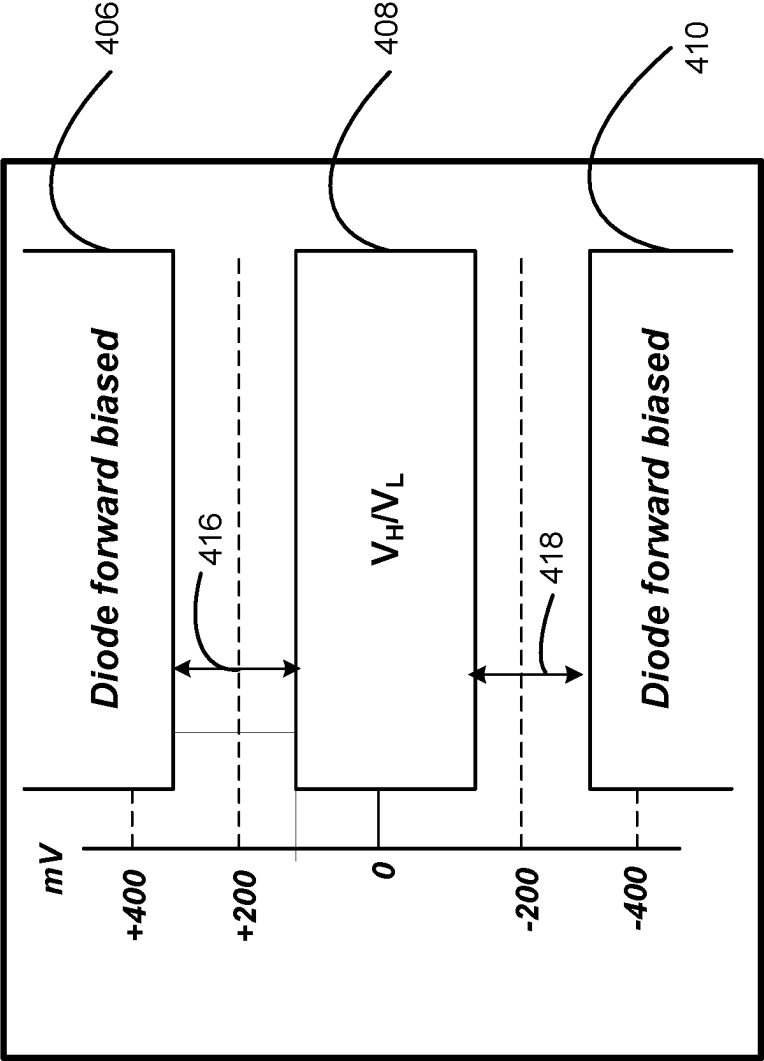


FIG. 4B

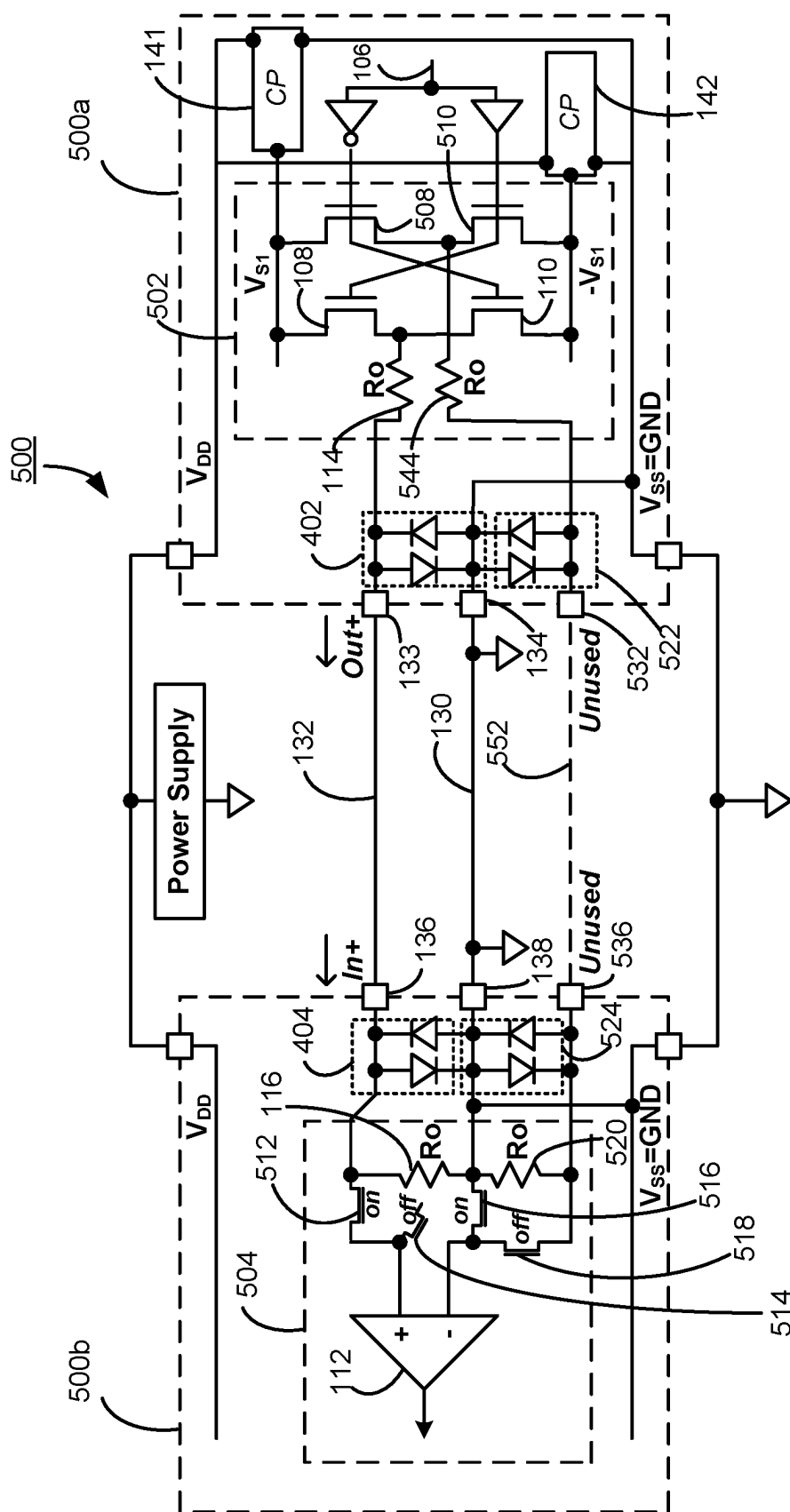


FIG. 5A

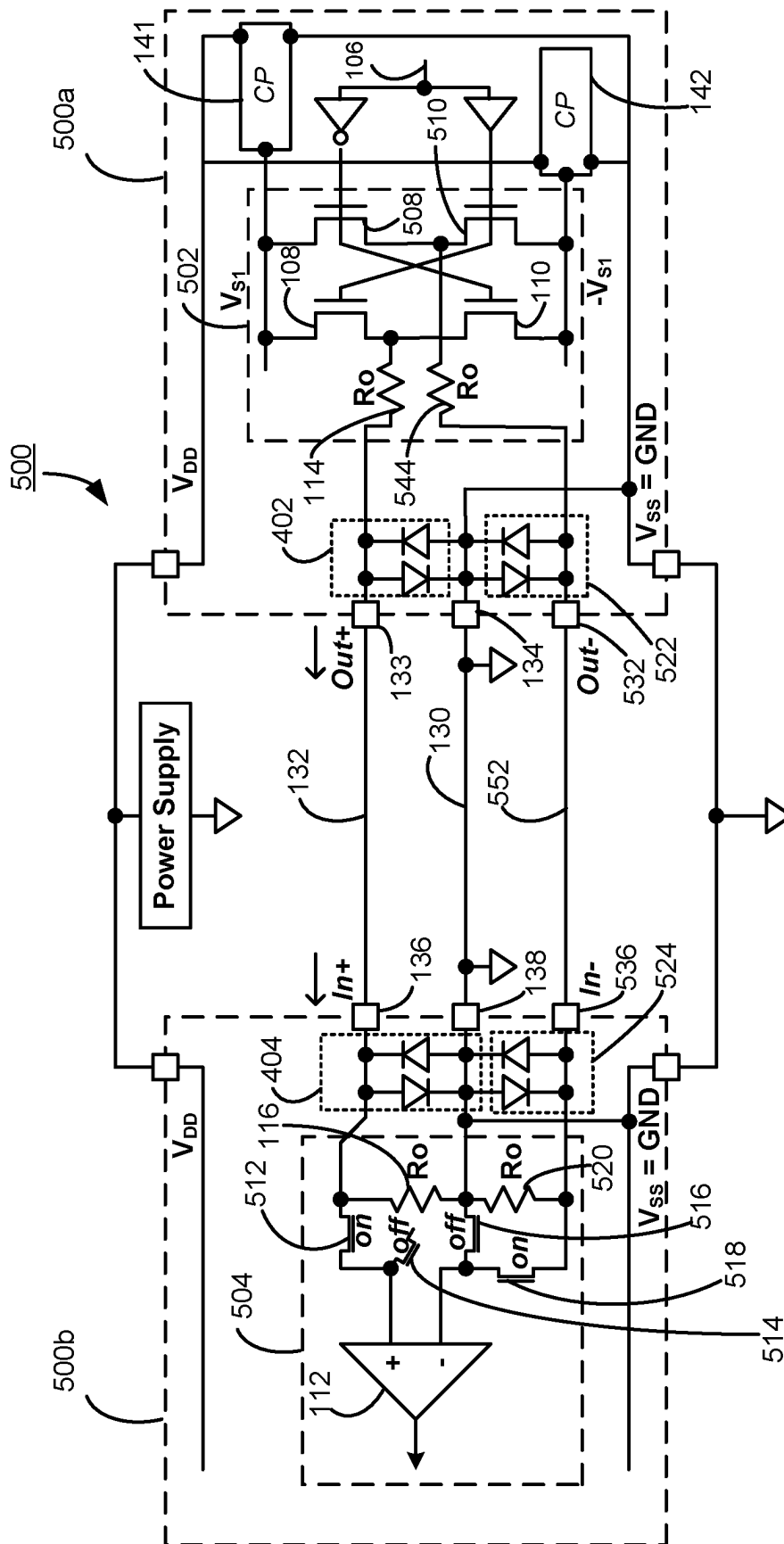


FIG. 5B

1

CONFIGURABLE, POWER SUPPLY VOLTAGE REFERENCED SINGLE-ENDED SIGNALING WITH ESD PROTECTION

BACKGROUND

The present disclosure relates to single-ended signaling.

Single-ended signaling is commonly used to transmit electrical signals over wires, such as those in contemporary memory systems. In single-ended signaling, a signal represented by a varying voltage is transmitted over a wire. The voltage varies above or below a reference voltage, which is used by a receiver of the signal to determine the digital values represented by the signal. In conventional single-ended signaling, the reference voltage is typically generated, either on-chip or off-chip, to be at the middle of a signal swing.

Conventional single-ended signaling typically has poor efficiency and limited speed. More specifically, conventional single-ended signaling has the disadvantage that (i) the reference voltage has to be generated for the receiver, (ii) it is difficult to match the reference voltage at the receiver with the reference voltage of the transmitter, (iii) noise such as simultaneous switching noise can be generated due to the use of a power supply voltage for signaling, and (iv) it is difficult to control a signal return path, resulting in crosstalk noise.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure herein is illustrated by way of example and without limitation in the figures of the accompanying drawings and in which like reference numerals refer to similar elements and in which:

FIGS. 1A, 1B, 1C, and 1D illustrate a data transmission system employing power supply voltage referenced single-ended signaling, according to one embodiment.

FIGS. 1E and 1F illustrate a cutaway view of an IC package including an integrated circuit of the data transmission system of FIGS. 1A through 1D on a printed circuit board.

FIGS. 2A, 2B, 2C, 2D, 2E, and 2F illustrate embodiments of signal voltage supplies employed by the data transmission system of FIGS. 1A through 1D.

FIGS. 3A and 3B illustrate receiver amplifiers suitable for use in the data transmission system of FIGS. 1A through 1D, according to one embodiment.

FIG. 3C is an alternative embodiment of a common-gate amplifier of FIG. 3B.

FIG. 4A illustrates a data transmission system employing ground-referenced single-ended signaling with ESD protection, according to one embodiment.

FIG. 4B illustrates a voltage swing of a signal in the data transmission system of FIG. 4A, according to one embodiment.

FIG. 5A illustrates a configurable data transmission system configured in a single-ended signaling mode, according to one embodiment.

FIG. 5B illustrates a configurable data transmission system configured in a differential signaling mode, according to one embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

The Figures and the following description relate to embodiments by way of illustration only. Alternative

2

embodiments of the structures and methods disclosed herein may be employed without departing from the principles of the present disclosure.

Embodiments of the present disclosure include a data transmission system using a single-ended signaling system in which a signaling voltage is referenced to a reference voltage that is a power supply voltage shared by a transmitter and a receiver. The signaling voltage swings above and below the reference voltage. The reference voltage may be ground voltage, generally taken to be the lowest power supply potential. The signaling voltage is derived from a first signal supply voltage and a second signal supply voltage, which may be generated by a pair of charge pumps that draw substantially the same amount of current from a power supply, regardless of the output drive level of the transmitter.

FIGS. 1A, 1B, 1C, and 1D illustrate a data transmission system **100** employing supply voltage referenced single-ended signaling, according to one embodiment. The data transmission system includes first and second integrated circuit devices (ICs) **100a** and **100b** coupled to each other via a plurality of conducting lines including signal line **132**. ICs **100a** and **100b** may be separate devices with separate IC packages, or combined within a die-stack or other multi-die package, or implemented by bare die. In the case of a multi-die package, the conducting lines may be implemented by intra-package conductors such as wire-bonds, flex conductors or any other suitable interconnection structures. In the case of separately packaged ICs or bare die, the conducting lines may be implemented partly by conductive structures for conveying signals (and establishing supply-rail connection) through the package to an external interconnect, and partly by printed circuit board (PCB) traces, PCB supply bus layers, flex cables or other conductive interconnects between the IC devices. With regard to system function, the ICs may be, for example and without limitation, a processor and chipset (or bridge or other application-specific IC (ASIC)), memory controller and memory device, or any other pair of ICs that carry out operations involving chip-to-chip signaling.

Still referring to FIG. 1A, IC device **100a** is to be connected to a power supply. IC device **100a** includes a transmitter **102** and terminals (such as pads or pins) **133**, **134**. Transmitter **102** is coupled between two signal supply voltages **V1** and **V2**, which are provided by one or more voltage sources **140**. In one embodiment, **V1** and **V2** are substantially symmetrical with respect to a power supply voltage V_p associated with the power supply (i.e., $V1 - V_p = V_p - V2$, at least approximately). Terminal **134** is coupled to a power supply rail **130** which is coupled to power supply voltage V_p , while terminal **133** is coupled to signal line **132**. By this arrangement, when input data **106** is applied to transmitter **102**, a signal that swings between about $\frac{1}{2}V1$ and about $\frac{1}{2}V2$ according to the input data **106** is output onto signal line **132** via terminal **133**. Thus, transmitter **102** forms a voltage mode transmitter that generates a data signal **120** having a signal voltage that swings substantially symmetrically about the power supply voltage V_p (i.e., between about $\frac{1}{2}V1$ and about $\frac{1}{2}V2$) to convey the bits of the input data **106**.

Because the transmitted signal swings symmetrically about the power supply voltage V_p , the power supply voltage may be used as a signaling reference voltage. In the embodiment shown, for example, the signal line **132** and the power supply rail **134** are coupled, via terminals **136** and **138**, to respective inputs of a receiver **104** in circuit **100b**. Receiver **104** amplifies the time-varying difference between the signal voltage and the power supply voltage to produce

the received data signal (RData) that corresponds to the originally transmitted data **106**. That is, receiver **104** compares the signal received via terminal **136** with the reference voltage supplied via terminal **138** to recover the transmitted signal and generate RData. For example, RData is “1” when the signaling voltage at terminal **136** is higher than the reference voltage at terminal **138**, and RData is “0” when the signaling voltage at terminal **136** is lower than the reference voltage at terminal **138**.

Bi-directional signaling may be effected by including in IC device **100b** a transmitter **102'** similar to transmitter **102** and in IC device **100a** a receiver **104'** similar to receiver **104**. The reverse-direction transmitter/receiver **102'/104'** may use the same signaling line **132** and/or the same supply-rail signaling reference **130** as transmitter/receiver **102/104** (thus driving signaling line **132** bi-directionally). Or, the reverse-direction transmitter/receiver **102'/104'** may drive a separate signal line, and/or employ a separate or different supply-rail signaling reference. Also, while not specifically shown within FIG. 1A, a mesochronous or plesiochronous timing arrangement may be used to enable synchronous data transmission and reception over link **132**. For example, a clock signal may be provided to both IC devices **100a** and **100b** (or forwarded from one device to the other) to provide timing for delivery of the input data **106** to the transmitter **102**, and to provide timing for latching of the output of receiver **104** within an input data register or like storage circuit. Alternatively (or additionally), a source synchronous timing reference (e.g., a strobe signal that transitions to mark the presence of data on the signaling line **132**) may be output from the transmitting IC to the receiving IC to control the signal sampling time within the receiving IC. Various timing vernier circuitry may be provided and calibrated to establish data output timing and/or data sample timing within desired margins.

In one embodiment, the V1/V2 signaling supply voltages are selected to effect a small signal swing V_s (e.g., $V_s \approx 1/2(V1 - V2)$) is substantially lower than corresponding on-chip logic levels) and thereby enable low-power signaling. The signaling supply voltages may be driven to programmably selected or designed setpoint voltages (as discussed below), or may be adaptively established or calibrated to achieve a threshold operating margin as determined by bit error rate or other signal quality metric.

Charge-pump-based voltage supplies or any other supply voltage circuits or voltage regulators may be used to implement the one or more voltage sources **140**, as discussed in more detail below. In one embodiment, for example, the voltage source(s) **140** is designed such that signal supply voltages are regulated and the net current drawn by transmitter **102** from the power supply remains substantially constant regardless of the output voltage level at terminal **133**, thereby reducing switching noise within the signaling system.

The signaling arrangement of FIG. 1A provides a number of benefits relative to conventional approaches. First, the power supply rail **130** may be implemented as a continuous conducting plane (e.g., power plane) that underlies all or most of the signal lines within an IC package or PCB, and thus constitutes a low and sometimes lowest impedance network in the signaling system **100**. Consequently, by coupling the reference output terminal **134** of transmitter **102** and the reference input terminal **138** of receiver **104** to this low impedance reference node (the power plane) **130**, a stable, consistent reference level may be ensured between various components of the signaling system, including transmitter **102** and receiver **104**.

Another advantage of signaling system **100** is that, as discussed below with reference to FIG. 1C, a signal current sourced by transmitter **102** flows over signaling line **132** to receiver **104** and then returns to transmitter **102** on power plane **130** via a path substantially parallel to the signaling line **132**. Thus, the return current may flow anti-parallel to the signal current, and the signal current and signal-return connections at the chip level are essentially differential. Consequently, noise and cross-talk that would otherwise be generated by return current flow that is not parallel to the signal current flow can be avoided. Altogether, the various benefits of the above-described signaling arrangement enable construction of a high-speed, ultra-low-power single-ended signaling system that requires on the order of 1 mW of signaling energy from the power supply—roughly 10-20 times lower than that of a conventional single-ended signaling system.

FIG. 1B illustrates an embodiment where a ground plane is used as power supply rail **130**, so that $V1 = +V_s$ and $V2 = -V_s$. Voltage source(s) **140** is shown as comprised of a voltage source **141** that provides $+V_s$ and a voltage source **142** that provides $-V_s$. Transmitter **102** may include signal-controlled switching elements **108**, **110** each coupled between a common drive node and respective signal supply voltages, $+V_s$ and $-V_s$. Transmitter **102** further includes a resistive element **114** coupled between the common drive node and terminal **133** to act as a series source termination (e.g., to match a characteristic impedance, Z_0 , of the signal line). Thus the transmitter's internal Z_0 impedance **114** and the impedance of the line **132**, terminated at Z_0 by termination impedance **116**, form a voltage divider that splits the two supply voltages, so that the signaling voltage of signal **120** on line **132** toggles between about $+V_s/2$ and $-V_s/2$, as shown in FIG. 1B. In some embodiments, the functions of switches **108**, **110** and resistor **114** may be combined in a single device, for example an MOS field effect transistor (MOSFET) operating in the linear or “triode” region of operation; in such embodiments, the linear relationship of the voltage between the drain and source terminals of the MOSFET and the current flowing between drain and source causes the device to operate in a mode approximating a resistor.

Still referring to FIG. 1B, receiver **104** may include a differential amplifier or other comparator circuit **112**. Amplifier **112** amplifies the time-varying difference between the ground rail and signal potentials to produce the received data signal (RData) that corresponds to the originally transmitted data **106**. As shown, a termination element (depicted as resistive element **116**) may be coupled between the input nodes **136** and **138** to terminate the incoming signal line according to the characteristic impedance, Z_0 .

Referring now to FIG. 1C, a signal current **150** sourced by transmitter **102** flows over signaling line **132** to receiver **104** and then returns to transmitter **102** on ground plane **130** via a path that can be made substantially parallel to the signaling line **132**. Thus, noise and cross-talk that would otherwise be generated by return current flow not parallel to the signal current flow can be reduced or eliminated.

In one embodiment, IC devices **100a** and **100b** include IC chips within respective semiconductor packages **184** each having an array of solder balls or other metal contacts that act as terminals for the IC devices. FIG. 1D illustrates IC devices **100a** and **100b** on a PCB **101**, according to one embodiment. FIG. 1D shows two signaling lines **180**, **182** between circuits **100a**, **100b**, as a subset of a plurality of signaling lines that would be present on the PCB **101** between circuits **100a** and **100b**. IC device **100a** includes a

5

plurality of transmitters, including transmitters **102a**, **102b**. IC device **100b** includes a plurality of receivers, including receivers **104a**, **104b**. IC devices **100a** and **100b** include a plurality of reference (GND) terminals **164**, **168** and **174**, **178**, respectively, that are coupled to a power plane (e.g., ground plane) **131** of the PCB **101**. The power plane **131** of the PCB **101** can be a solid metal layer in the PCB **101**, a metal layer with windows or holes, or it can include a plurality of conducting lines each coupled to a same power supply voltage. Although FIG. 1D shows that power plane **131** is between IC devices **100a** and **100b**, in practice, power plane **131** may extend under IC devices **100a** and **100b**.

IC devices **100a** and **100b** further include a plurality of signaling terminals **162**, **166** and **170**, **174**, respectively. To facilitate routing of the signaling lines, the solder balls acting as or coupled to the signaling terminals **162**, **166** and **170**, **174** can be placed closer to the edge of the respective semiconductor packages **184** than the solder balls acting as or coupled to the reference (e.g., GND) terminals **164**, **172** and **168**, **178**. For example, in the IC packaging **184** of circuit **100b** (as shown by the insert in FIG. 1D), the signaling terminals **170**, **174** are placed as outer balls of the IC packaging **184** to facilitate easy connection of the balls to the signaling lines **180** and **182** outside the IC package. On the other hand, the reference terminals **172**, **174** can be connected to the inner balls of the IC packaging **184**, because they only need to be coupled to a power plane (e.g., GND plane) within the package, which is coupled to the power plane **131** of the PCB **101**, to which the package is soldered or otherwise attached. Such ground package pins **172**, **178** are not as expensive as signal pins **170**, **174**, since they are connected directly through vias to a solid conducting plane, thereby requiring no trace routing. Also, although FIG. 1D illustrates that separate reference (e.g., GND) terminals **164**, **172** and **168**, **178** are used for each of the corresponding signaling terminals **162**, **170** and **166**, **174**, it is also possible for two or more of the signal pins **162**, **166** and **170**, **174** to share a common reference terminal, thereby further reducing the number of package pins needed for the reference pins.

FIG. 1E illustrates a cutaway view of an IC package including IC device **100a** (or IC device **100b**) on a printed circuit board. According to one embodiment, referring to FIG. 1E, IC device **100a** is mounted on a laminated IC package substrate **200**. Package substrate **200** is a multi-layer package substrate having signal wiring conductive layers, including a top layer (e.g., signal trace **202**) and a bottom layer (e.g., solder-ball pads **206a** and **206b**). Package substrate may also include additional conducting layers separated from each other and from the top and bottom layers by insulating layers **204**. The additional layers may be used for power delivery and signal return current conduction. For example, layer **201** in FIG. 1E may be a signal return (and/or reference) plane for the previously described single-ended signaling system, while layer **205** may be a plane used to connect to some other power supply voltage. In one embodiment, reference plane **201** corresponds to the ground reference plane **130** of FIG. 1B (i.e., coupled to an external ground reference) and is sometimes referred to as ground plane **201** below. In alternative embodiments, reference plane **201** may be coupled to another DC reference or even a time-varying reference.

Still referring to FIG. 1E, integrated circuit package **200** may be attached both electrically and mechanically to printed wiring board **190** via an array of solder balls, including solder balls **207a** and **207b**. Printed wiring board **200** may be constructed similarly or identically to laminated

6

package substrate **200**, with surface conducting layers, such as layer **191**, inner conducting planes, such as plane **194**, and perhaps additional internal signal layers and power supply planes, that are separated from each other by insulating layers **193**.

Still referring to FIG. 1E, IC device **100a** includes one or more instances of ground-referenced signaling receivers and transmitters, such as **102** and **104** in FIG. 1A. The output pads of these units (corresponding to **133** and **134** in FIG. 1A) are connected to conducting traces and planes on package **200** via wire bonds **188** and **189**, respectively. In FIG. 1E, the signal terminal **133** is connected to package trace **202** via wire-bond **188**. Signal current, denoted by the dashed lines, flows from the chip **100a**, through wire-bond **188** to package trace **202**, thence to through-package via **203**, where it is carried to the bottom layer of the package to solder-ball pad **206a**. Signal current further flows through solder ball **207a** to printed wiring board solder-ball pad and signal trace **191**, and from there to another packaged integrated circuit such as IC device **100b**. Return current (also shown by a dashed line) flows back from the second packaged integrated circuit over a power supply plane **194**, to via **195**, where it is conducted up to solder-ball pad **192**, through solder ball **207b**, through solder-ball pad **206b** and via **208** to the on-package conducting ground plane **201**. The electromagnetic interaction between the signal current and this return current will tend to cause the return current to flow in plane **201** in a manner anti-parallel to the signal current in signal trace **202** (i.e., reverse-direction current flow in a path parallel to the signal trace **202**), finally reaching integrated circuit **100a** through wire bond **189**. In one embodiment, the solder balls **207a/207b**, solder-ball pads **206a/206b**, and through-package vias **203**, **208** that carry the signal current and return current are disposed in close proximity. The close proximity of the solder balls, solder-ball pads, and through-package vias that carry the signal current and return current help avoid large series inductances from accumulating in the overall signal path, and thus further reduces or eliminates cross-talk with neighboring signals and their return currents.

FIG. 1E is shown merely by way of example, as there are many other ways of implementing an integrated circuit package and printed wiring boards, including, but not limited to “flip-chip” (C4) construction, package-on-package (POP), and other packaging constructions. Although the technology used and the configuration of the package may vary, the following features should preferably be maintained for optimal performance: (1) the use of a common, low-impedance power supply conducting plane (e.g., power supply conducting plane) that carries all return currents from one or more supply-reference single-ended signaling components (i.e., transmitters and receivers); and (2) provision for a separate path from the conducting plane to the reference terminals of both transmitters and receivers (e.g., pins **134** and **138** in FIG. 1A), which path may include, for example and without limitation, board and package conducting planes, solder balls, through-package and through-board vias, wire bonds, and C4 solder bumps. Furthermore, to the extent permitted by the packaging construction technique, the signaling system may also include an arrangement that forces the signal return current to flow anti-parallel to the corresponding signal current, while not overlapping with the signal current return path of other signaling components.

FIG. 1F illustrates yet another packaging embodiment, similar to the embodiment of FIG. 1D (and including all the variants mentioned above), but in which the ground plane

7

201 extends beneath the IC die **100a**. In the embodiment shown, the IC die **100a** is insulated from the ground plane by a relatively thin layer of epoxy **197**. Other insulating materials may be used in alternative embodiments. This approach generally provides all the benefits of the arrangement of FIG. 1E while maintaining the presence of the ground plane **201** beneath the IC die and thus providing potentially superior shielding and noise suppression.

FIG. 2A illustrates one embodiment of the voltage source(s) **140**. As shown in FIG. 2A, voltage source(s) **140** includes charge pumps **212** and **214** for generating the signal supply voltages $+V_s$ and $-V_s$, respectively, and a charge pump regulator **210**. Regulator **210** is coupled to the signal supply voltages $+V_s$ and $-V_s$, and receives a reference voltage V_{ref} that represents a desired voltage level for V_s . It controls or regulates the charge pumps **212** and **214** using control signals ϕ_{2b} and ϕ_{2a} , respectively. Charge pumps **212** and **214** are each coupled between first and second power supply voltages (e.g., V_{dd} and GND), and are configured to draw current from the power supply to provide the signal supply voltages $+V_s$ and $-V_s$, and to maintain the signal supply voltages $+V_s$ and $-V_s$ at desired levels in response to control signal ϕ_1 and to control signals ϕ_{2b} and ϕ_{2a} , respectively.

FIG. 2B illustrates an embodiment of regulator **210**. As shown in FIG. 2B, regulator **210** receives a pair of non-overlapping clocks ϕ_1 and ϕ_2 (or, it may include a clock signal source **220** that generates the pair of non-overlapping clocks ϕ_1 and ϕ_2). A pair of equally sized resistors **216** and **218** divides a voltage difference between $+V_s$ and $-V_s$ to produce a voltage V_{cm} (a common-mode voltage between the positive and negative signal supply voltages). The signal supply voltages $+V_s$ and $-V_s$ are nominally symmetric about GND. A digital or “bang-bang” control loop including sense amplifier/latch **224** amplifies the difference (error) voltage between V_{cm} and GND. When the voltage on V_{cm} is higher than the voltage on GND, a signal on line **228** is asserted, and during the following ϕ_2 interval, AND gate **234** outputs signal ϕ_{2a} that tracks signal ϕ_2 (i.e., signal ϕ_{2a} is asserted if signal ϕ_2 is asserted). On the other hand, if the voltage on V_{cm} is lower than the voltage on GND, sense amplifier/latch **224** amplifies this difference and de-asserts the signal on line **228**, such that ϕ_{2a} is not asserted.

Likewise, digital or “bang-bang” control loop including sense amplifier/latch **222** that amplifies the difference (error) voltage between $+V_s$ and V_{ref} . If the voltage on $+V_s$ is lower than the voltage V_{ref} , a signal on line **226** is asserted, and during the following ϕ_2 interval, AND gate **232** outputs signal ϕ_{2b} that tracks signal ϕ_2 (i.e., signal ϕ_{2b} is asserted if signal ϕ_2 is asserted). On the other hand, if the voltage $+V_s$ is higher than the voltage V_{ref} , sense amplifier/latch **222** amplifies this difference and de-asserts the signal on line **226**, such that ϕ_{2b} is not asserted. By asserting and de-asserting ϕ_{2a} and ϕ_{2b} based on comparison of V_{cm} with GND and $+V_s$ with V_{ref} , respectively, the signal supply voltages $+V_s$ and $-V_s$ can be maintained at the desired levels, as discussed below.

FIG. 2C illustrates another embodiment of regulator **210**. Here, the signal supply voltage $+V_s$ is fed back into comparator **242** for comparison with the reference voltage V_{ref} . The output **246** of comparator **242** is asserted if $+V_s$ is smaller than V_{ref} . Thus, the output ϕ_{2b} of AND gate **252** follows the clock signal ϕ_2 when $+V_s$ is lower than V_{ref} and is not asserted otherwise. Likewise, the signal supply voltage $-V_s$ is fed back into window comparator **244** for comparison with the reference voltage V_{ref} . The output **248** of window comparator **244** is asserted if the difference

8

between GND and $-V_s$ is smaller than the difference between V_{ref} and ground (i.e., $GND - (-V_s) < V_{ref} - GND$, or V_s is smaller than V_{ref}). Thus, the output ϕ_{2a} of AND gate **254** follows the clock signal ϕ_2 when the difference between GND and $-V_s$ is smaller than the difference between V_{ref} and GND. Otherwise, it is not asserted.

FIGS. 2D and 2E illustrate embodiments of charge pumps **212** and **214**, respectively. Referring to FIG. 2D, charge pump **212** includes a switching device **262** (shown as a P-type metal-on-semiconductor (MOS) field effect transistor (PFET)) that is turned on or off according to clock signal ϕ_1 , and a switching device **264** (shown as an N-type MOS field effect transistor (NFET)) that is turned on or off according to clock signal ϕ_{2b} . Charge pump **212** further includes a capacitor **266** and a capacitor **268**. Capacitor **266** has a much smaller capacitance than that of capacitor **268** and is therefore sometimes referred to as C_{small} , while capacitor **268** is sometimes referred to as C_{big} . C_{big} **268** is coupled between $+V_s$ and GND, so the voltage across capacitor **268** is the signal supply voltage $+V_s$.

FIG. 2F is timing chart showing signals ϕ_1 , ϕ_2 , ϕ_{2b} , the voltage across capacitor **266** ($V_{C_{small}}$), and the voltage across capacitor **268** ($+V_s$) in comparison with V_{ref} during a series of time periods after IC device including the charge pumps is turned on. As shown in FIG. 2F, during time period t_1 , when ϕ_1 is asserted, switch **262** is turned on, allowing charge to flow from V_{dd} to capacitor **266**, so that a voltage across capacitor **266**, $V_{C_{small}}$, increases to a level between V_{dd} and GND. The voltage across C_{big} is lower than V_{ref} so ϕ_{2b} is tracking ϕ_2 . During time period t_2 , when ϕ_1 is de-asserted while ϕ_2 and thus ϕ_{2b} is asserted, switch **262** is turned off while switch **264** is turned on, so that stored charge in C_{small} is shared with C_{big} and the voltage across capacitor **268**, $+V_s$, increases while $V_{C_{small}}$ decreases.

Since the capacitance of C_{small} is much smaller than the capacitance of C_{big} , and the increase in voltage across C_{big} during time period t_2 is much smaller than the decrease in voltage across C_{small} , so the voltage across C_{big} at the end of time period t_2 is slightly higher than the voltage across C_{big} at the start of time period t_2 . The above charging of C_{small} and subsequent charging of C_{big} may repeat in response to signals ϕ_1 and ϕ_{2b} , until the signal supply voltage $+V_s$ reaches or is higher than V_{ref} at the end of time period t_3 . As discussed above, as long as $+V_s$ remains at or above V_{ref} during time periods t_4 and t_4 , ϕ_{2b} is not asserted, meaning switch **264** does not turn on and C_{big} does not get further charged by C_{small} . When $+V_s$ drops below V_{ref} during time period t_6 , ϕ_{2b} follows ϕ_2 again in time period t_7 , and charges from C_{small} will be shared with C_{big} until $+V_s$ is higher than V_{ref} again.

The signal and voltage curves shown in FIG. 2F are for illustrative purposes only and are not to replicate or scale with signals or voltages in a real IC device. In some embodiments, the voltage between V_{dd} and GND is much higher than V_{ref} . For example, V_{dd} may be about 1 volt, while V_{ref} is about 0.1 volt. In such case, if the charge pumps run at a frequency of, for example, 1 GHz, data is being driven out at, for example, 4 Gbps, and the impedance of transmitter **102** (thus the impedance of the load) is, for example, 50 ohms, C_{small} can be about 0.75 pF to about 1.25 pF while C_{big} can be about 50 pF to about 250 pF.

On the other hand, charge pump **214** includes switches **272**, **282** that are turned on or off according to clock signal ϕ_1 , and switches **274**, **284** that are turned on or off according to clock signal ϕ_{2a} . Charge pump **214** further includes a capacitor C_{small} **276** and a capacitor C_{big} **278**. C_{big} **278** is coupled between GND and $-V_s$ so the voltage across C_{big}

278 is the signal supply voltage $-V_s$. Switches 272, 282 are turned on when clock signal ϕ_1 is asserted so that charge is extracted from power supply V_{dd} via switch 272 and 282 and stored in capacitor C_{small} 212. Switches 274 and 284 are turned on when signal ϕ_{2a} is asserted and signal ϕ_1 is de-asserted, so charges stored in C_{small} is pumped into Cbig. This process repeats until the common mode voltage V_{cm} is lower than GND, or the difference between GND and V_s is larger than the difference between V_{ref} and GND. As discussed above in association with FIGS. 2B and 2C, as long as the common mode voltage V_{cm} remains lower than GND, or the difference between GND and V_s remains larger than the difference between V_{ref} and GND, ϕ_{2a} remains de-asserted and no charge is pumped from C_{small} 276 to Cbig 278 to further increase the voltage across Cbig 278. When the common mode voltage V_{cm} becomes higher than GND, or the difference between GND and V_s becomes smaller than V_{ref} , ϕ_{2a} follows ϕ_2 again and Cbig 278 gets charged from C_{small} until V_{cm} becomes lower than GND, or the difference between GND and V_s becomes larger than the difference between V_{ref} and GND again.

FIGS. 2D and 2E shows that switches 262 and 272 are implemented using P-type metal-oxide-semiconductor field effect transistors (P-MOSFET or PFET), while switches 264, 274, 282 and 284 are implemented using N-type MOSFETs (NFET). PFET 262 or 272 is turned on when ϕ_1 is asserted by virtue of inverter 260 or 270, respectively, which inverts the sense of ϕ_1 to drive the gate terminal of PFET 262 or 272. Those skilled in the art will easily understand that other types of switches can be used in place of the MOSFETs. The capacitors 266, 268, 276, and 278 may be implemented using MOS capacitors, but other types of capacitors can alternatively be employed.

The embodiments of FIGS. 2A through 2F are shown by way of examples. There are many other possible embodiments of on-chip voltage sources or regulators. For example, some embodiments may have additional C_{small} capacitors and additional switches that charge these C_{small} capacitors in a series circuit arrangement and discharge the C_{small} capacitors in a parallel arrangement, thereby providing higher overall efficiency than that of the simple charge pumps of FIGS. 2D and 2E. In a further example, instead of using clocks to gate the MOSFETs on and off in the charge pumps, other type of switches that respond to other characteristics of control signals such as frequency, amplitude or duty cycle of the control signals may be used. In yet another example, switched capacitors may be used in place of the switch and capacitor combinations in the charge pumps. In still other examples, other regulating means such as a linear regulator or switching regulator may be used to control the signal supply voltages.

In one embodiment, charge pumps 212 and 214 are implemented with components having substantially the same efficiency. In the embodiment shown in FIG. 2A, charge pump 212 draws current from V_{dd} when transmitter 102 is transmitting "1"s while charge pump 214 draws current from V_{dd} when transmitter 102 is transmitting "0"s. Thus, assuming that the transmitter 102 has matching impedance with the transmission lines 132, 130, the transmitter 102 draws approximately the same amount of supply current from the power supply regardless of which way the transmitter 102 is driving the current (i.e., sourcing current from transmitter 102 to receiver 104, or sinking current from receiver 104 to transmitter 102). Thus, to the extent that charge pumps 212 and 214 have about the same efficiency, the complementary configuration of the voltage source(s) 140 ensures that supply current variation in signaling system

100 is near zero, thereby reducing or eliminating simultaneous switching noise (SSO) inherent in conventional single-ended signaling schemes.

FIGS. 3A, 3B, and 3C illustrate several alternative embodiments for the receiver amplifiers used in the data transmission system 100 of FIGS. 1A, through 1D, when the GND is chosen as the reference voltage, according to one embodiment. FIG. 3A illustrates a PMOS amplifier 302 that can be used in some embodiments of receiver 104, coupled to the two terminals 136, 138 corresponding to the signaling voltage and GND reference voltage, respectively. This amplifier is a conventional PMOS differential amplifier with resistor loads. Numerous alternative embodiments of a PMOS differential amplifier are possible. FIG. 3B illustrates one embodiment of a common gate amplifier 304 that can be used in receiver 104 in other embodiments, coupled to the two terminals 136, 138 corresponding to the signaling voltage and GND reference voltage, respectively. The common gate amplifier 404 in FIG. 3B is implemented with NMOS transistors and thus may have a higher gain-bandwidth product than the PMOS amplifier 402 of FIG. 3A.

FIG. 3C is a common-gate amplifier 306 according to another alternative embodiment. Current source 906 establishes a reference current in NFET 914, thereby generating gate reference voltage V_{cas} . NFETs 910 and 912, which are the amplifying transconductances in amplifier 306, are drawn at the same shape factor (width/length) as 914 so that they carry about the same current as 914, when their inputs, driven into their source terminals, attached to pins 136 and 138, are at substantially the same potential, since in this case the gate-to-source voltages (V_{gs}) for all three transistors are substantially equal and the shape factors are identical. The data signal, attached to pin 136, is compared in amplifier 306 with the signal reference voltage, input on pin 138 from the power supply GND plane 130. The current from line to GND develops a voltage across termination resistor 920; this voltage swings from about $+V_s/2$ to $-V_s/2$. When the voltage on line (pin 136) is higher than the voltage on GND (pin 138) NFET 910's V_{gs} is less than V_{gs} at NFET 912. Therefore, less current flows through 910 and through 912, and the voltage drop through load resistor 902 is smaller than the voltage drop through load resistor 904. This causes the voltage on outH to rise above the voltage on outL. On the other hand, if the voltage on 136 is lower than the voltage on 138, then NFET 910's V_{gs} is greater than V_{gs} for NFET 912. Consequently NFET 910 carries more current than 912, and the voltage across 902 is larger than the voltage across 904. In this case, the voltage on outH is less than the voltage on outL. Under these circumstances the current driven back into line 132 (pin 136) by NFET 910 is larger than the current driven into the line 132 by 910 when the voltage on line exceeds the voltage on GND. To prevent this current unbalance from introducing an unwanted voltage offset in amplifier 306, a compensating resistor 922, R_C , is used to return some of the current to the negative power supply $-V_s$.

Still referring to FIG. 3C, the magnitude of the voltage between outH and outL is higher than the magnitude of the voltage between line (pin 136) and GND (pin 138) thanks to the gain of amplifier 306; the gain of 306 is about $G_m \times R_L$, where G_m is the transconductance of NFETs 910 and 912, and R_L is the resistance of load resistances 902 and 904.

Still referring to FIG. 3C, it will be clear to those skilled in the art, that the terminating impedance presented across pins 136 and 138 is the parallel combination of the terminating resistor R_T 920, the compensating resistor R_C 922, and the input impedance of the amplifier, which is about $1/G_m$, where G_m is the transconductance of transistors 910

11

and 912. This parallel combination of resistances should be adjusted to about the characteristic impedance of the transmission line formed by line 132 and GND to avoid reflections in the transmission line.

The foregoing embodiments of input amplifiers are shown by way of examples. Many alternative embodiments of input amplifiers can be employed in receiver 104 or 104' in system 100.

FIG. 4A illustrate a data transmission system employing GND referenced single-ended signaling with ESD protection, according to one embodiment. The data transmission system 400 includes circuit 100a including the transmitter 102 and circuit 100b including the receiver 104, and employs the single-ended signaling scheme as described above using the signaling line 132 and GND reference line 130. Voltage sources such as charge pumps (CP) 141, 142 supply the positive and negative signal supply voltages V_{S1} and $-V_{S1}$, and either switch 108 or switch 110 is turned on to source or sink line current on the signaling line 132 responsive to input data 106, as explained above. The signaling voltage is developed across resistor 116 and detected by input amplifier 112 to recover the input data 106.

The data transmission system 400 additionally includes ESD (Electrostatic Discharge) protection devices 402, 404. In one embodiment, each ESD protection device 402, 404 is comprised of a pair of cross-coupled diodes connected between the signaling voltage 132 and the GND reference voltage 130 of the single-ended signaling system, with ESD protection device 402 connected on the transmitter side 102 and the ESD protection device 404 connected on the receiver side 104. Note that the cross-coupled diodes in ESD protection device 404 are coupled between the inputs of the input amplifier 112 in parallel with resistor 116. The ESD protection devices 402, 404 have the benefit of reducing the input capacitance of the I/O pins 133, 134 and 136, 138 and protecting the pins 133, 134 and 136, 138 from electrostatic discharge. Also, as shown in FIG. 4B, the allowed voltage swing 408 ($V_L = -50$ mV to $V_H = +50$ mV) of the signaling voltage 132 should be within the margins 416, 418 between the regions 406, 408 in which diodes 402, 404 become forward biased. Symmetrically connecting the diodes 402, 404 to the GND signaling reference voltage 130 allows the voltage swing of the signaling voltage on line 132 to be larger than for asymmetric connection, because the margins 416, 418 between the regions 406, 410 also become symmetrical with respect to the GND reference voltage 130 and thus no margins 416, 418 between the two regions 406, 410 are wasted.

FIG. 5A illustrates a configurable data transmission system configured in single-ended signaling mode, according to one embodiment. The data transmission system 500 includes circuits 500a, 500b. Circuit 500a includes a transmitter 502 and circuit 500b includes a receiver 504. Transmitter 502 and receiver 504 and employs the ground-referenced single-ended signaling as described above using the signaling line 132 and GND reference line 130. Charge pumps (CP) 140, 142 supply the positive and negative signal supply voltages V_{S1} and $-V_{S1}$ for generating the signaling voltage on line 132, and either switch 108 or switch 110 is turned on to source or sink line current on the signaling line 132 responsive to input data 106 as explained above. The signaling voltage is detected across resistor 116 and detected by input amplifier 112 to recover the input data 106. ESD protection devices 402, 404 each comprised of a pair of cross-coupled diodes are also connected between the terminals 133, 134 and between the terminals 136, 138, respectively.

12

Circuit 500a additionally includes terminal 532 and ESD protection device 522, and transmitter 502 additionally includes switches 508, 510 and resistor 544. Circuit 500b additionally includes ESD protection device 524 and terminal 536, and the receiver 504 additionally includes switches 512, 514, 516, 518. An additional transmission line 552 is coupled between terminal 532 and terminal 536. Terminals 133, 134, 532 form one configurable unit (set) of pins of the transmitter 502 among a number of output pins that would be present on the IC including the transmitter 502. Similarly, terminals 136, 138, 536 form one configurable unit (set) of pins of the receiver 504 among a number of output pins that would be present on the IC including the receiver 504. However, when the data transmission system 500 is used in single-ended signaling mode, terminals 532, 536, line 552, ESD protection devices 522, 524 are not used, because switch 518 is turned off, disconnecting the input amplifier 112 from terminal 536 and line 552. Also, switch 512 and switch 518 are turned on, connecting terminals 136 and 138 to the inputs of amplifier 112. Switch 514 is a dummy transistor for providing matched loading corresponding to switch 518, but is also turned off. On the transmitter side 502, switches 508, 510 and resistor 544 are not used because they are connected to line 552 that is disconnected from input amplifier 512. Thus, the data transmission system 500 is configured for single-ended signaling mode, similar to the transmission system 400 of FIG. 4, using signaling line 132 and the GND reference line 130.

FIG. 5B illustrates a configurable data transmission system configured in differential signaling mode, according to one embodiment. The components and structure of the data transmission system 500 shown in FIG. 5B are the same as those of the data transmission system 500 shown in FIG. 5A. However, the data transmission system 500 in FIG. 5B is configured to operate in differential signaling mode. Differential signaling transmits data by means of two complementary signals sent on two separate wires. Here, lines 132 and 552 are used as the complementary (positive and negative) signaling lines for differential signaling, and the GND reference line 130 is not used. On the receiver side 504, switches 512, 518 are turned on to connect terminals 136, 536 (and signaling lines 132, 552) to input amplifier 112, respectively, but switch 516 is turned off to disconnect terminal 138 (and the GND reference line 130) from the input amplifier 112. Switch 514 is also turned off. On the transmitter side 502, both pairs of switches 108, 110 and 508, 510 are used to generate the complementary differential signals to be sent over transmission lines 132, 552. Switches 108, 110 are connected to signaling line 132 via resistor 114 to generate the positive differential signal (Out+) on line 132, and switches 508, 510 are connected to signaling line 552 via resistor 544 to generate the negative differential signal (Out-). Further, both ESD protection devices 402, 522 are used on the transmitter side 502 and both ESD protection devices 404, 524 are used on the receiver side 504. Thus, the data transmission system 500 is configured for differential signaling mode, using both signaling lines 132, 552 and terminals 133, 532 on transmitter side 502 and terminals 136, 536 on the receiver side 504.

The data transmission system described in FIGS. 5A and 5B is configurable to a single-ended signaling system or a differential signaling system in a convenient manner, simply by turning on or off switches 516, 518. For example, by adding a pin (e.g., pin 536) for every pair of pins (pins 136, 138) on the receiver side 504, a signaling interface that is configurable and versatile with high signal integrity can be achieved.

13

Alternatively, no additional pins are required. In single-ended signaling mode each pair of pins are line pin and reference or GND pin, respectively. In differential signaling mode they are lineP pin and lineN pin, respectively. The lineN half of the transmitter is turned off in single-ended signaling mode, and a large NFET switch can connect what was lineN to internal Vss. A dummy switch on lineP can be added to avoid imbalances. Both terminals would have cross-coupled diodes returned to the GND rail. At the receiver, no switches are required. The lineN input can just be connected to GND. Thus, a configurable signal transmission system similar to that described in FIGS. 5A and 5B but without the extra pins 536 and 532 can be formed.

It should be noted that the various integrated circuits, dice and packages disclosed herein may be described using computer aided design tools and expressed (or represented), as data and/or instructions embodied in various computer-readable media, in terms of their behavioral, register transfer, logic component, transistor layout geometries, and/or other characteristics.

When received within a computer system via one or more computer-readable media, such data and/or instruction-based expressions of the above described circuits may be processed by a processing entity (e.g., one or more processors) within the computer system in conjunction with execution of one or more other computer programs including, without limitation, net-list generation programs, place and route programs and the like, to generate a representation or image of a physical manifestation of such circuits. Such representation or image may thereafter be used in device fabrication, for example, by enabling generation of one or more masks that are used to form various components of the circuits in a device fabrication process.

In the foregoing description and in the accompanying drawings, specific terminology and drawing symbols have been set forth to provide a thorough understanding of the present invention. In some instances, the terminology and symbols may imply specific details that are not required to practice the invention. For example, any of the specific numbers of bits, signal path widths, signaling or operating frequencies, component circuits or devices and the like may be different from those described above in alternative embodiments. In other instances, well-known circuits and devices are shown in block diagram form to avoid obscuring the present invention unnecessarily. Additionally, the interconnection between circuit elements or blocks may be shown as buses or as single signal lines. Each of the buses may alternatively be a single signal line, and each of the single signal lines may alternatively be buses. Signals and signaling paths shown or described as being single-ended may also be differential, and vice-versa. A signal driving circuit is said to "output" a signal to a signal receiving circuit when the signal driving circuit asserts (or deasserts, if explicitly stated or indicated by context) the signal on a signal line coupled between the signal driving and signal receiving circuits. The term "coupled" is used herein to express a direct connection as well as a connection through one or more intervening circuits or structures. Integrated circuit device "programming" may include, for example and without limitation, loading a control value into a register or other storage circuit within the device in response to a host instruction and thus controlling an operational aspect of the device, establishing a device configuration or controlling an operational aspect of the device through a one-time programming operation (e.g., blowing fuses within a configuration circuit during device production), and/or connecting one or more selected pins or other contact structures of the

14

device to reference voltage lines (also referred to as strapping) to establish a particular device configuration or operation aspect of the device. The terms "exemplary" and "embodiment" are used to express an example, not a preference or requirement.

While the invention has been described with reference to specific embodiments thereof, various modifications and changes may be made thereto without departing from the broader spirit and scope. For example, features or aspects of any of the embodiments may be applied, at least where practicable, in combination with any other of the embodiments or in place of counterpart features or aspects thereof. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

What is claimed is:

1. An integrated circuit, comprising:

a first power supply terminal to receive a first power supply voltage;

a second power supply terminal to receive a second power supply voltage;

a first terminal;

a second terminal to be at the second power supply voltage;

a transmitter to transmit a transmit signal via the first terminal, the transmit signal representing a first data stream using a first voltage level and a second voltage level, the first voltage level and the second voltage level to be substantially symmetric above and below the second power supply voltage; and

a receiver to receive a second data stream that uses the first voltage level and the second voltage level, the receiver configurable to operate in a single-ended signaling mode where a single-ended signal is received via the first terminal, and a differential-signaling mode in which a differential signal is received via the first terminal and a third terminal.

2. The integrated circuit of claim 1, further comprising:

a first electrostatic discharge protection circuit couple between the first terminal and the second terminal and having two non-overlapping forward bias regions.

3. The integrated circuit of claim 2, wherein the first voltage level and the second voltage level are between the two non-overlapping forward bias regions.

4. The integrated circuit of claim 1, wherein the second terminal is part of a signal return path for the transmit signal.

5. The integrated circuit of claim 1, wherein the second power supply voltage is a ground voltage.

6. The integrated circuit of claim 1, wherein the second power supply voltage is other than a ground voltage.

7. The integrated circuit of claim 1, wherein the receiver comprises a differential amplifier to receive the second data stream.

8. An integrated circuit, comprising:

a first terminal;

a second terminal to be at a first power supply voltage;

a transmitter circuit to transmit a first signal while a first charge pump and a second charge pump draw a total power supply current from a second power supply voltage that is substantially independent of a voltage level of the first signal, the first signal switching between a first voltage level and a second voltage level, the first voltage level and the second voltage level being substantially symmetrical with respect to the first power supply voltage; and

a receiver circuit to receive a second signal that switches between the first voltage level and the second voltage level, the receiver circuit configurable to operate in a

15

single-ended signaling mode where a single-ended signal is received via the first terminal, and a differential-signaling mode in which a differential signal is received via the first terminal and a third terminal.

9. The integrated circuit of claim 8, wherein the receiver circuit comprises a differential amplifier to receive the second signal.

10. The integrated circuit of claim 8, further comprising: a first charge pump to draw a first current from a second power supply voltage to generate the first voltage level; and

a second charge pump to draw a second current from the second power supply voltage to generate the second voltage level.

11. The integrated circuit of claim 10, wherein the first power supply voltage is a ground voltage.

12. The integrated circuit of claim 10, wherein the first power supply voltage is other than a ground voltage.

13. The integrated circuit of claim 8, further comprising: a first electrostatic discharge protection circuit couple between the first terminal and the second terminal and having two non-overlapping forward bias regions.

14. The integrated circuit of claim 13, wherein the first voltage level and the second voltage level are between the two non-overlapping forward bias regions.

15. A method, comprising:

receiving, at a first power supply terminal, a first power supply voltage;

receiving, at a second power supply terminal, a second power supply voltage;

16

transmitting, via a first terminal, a transmit signal representing a first data stream using a first voltage level and a second voltage level, the first voltage level and the second voltage level being substantially symmetric above and below the second power supply voltage; and configuring a receiver in one of a first mode and a second mode to receive a second data stream that uses the first voltage level and the second voltage level, the first mode to receive the second data stream via the first terminal, the second mode to receive the second data stream via the first terminal and a third terminal.

16. The method of claim 15, wherein the first mode configures the receiver to operate in a single-ended signaling mode where a single-ended signal is received via the first terminal.

17. The method of claim 16, wherein the receiver comprises a differential amplifier and, in the first mode, the first terminal is coupled to the differential amplifier to receive the second data stream.

18. The method of claim 15, wherein the second mode configures the receiver to operate in a differential-signaling mode in which a differential signal is received via the first terminal and the third terminal.

19. The method of claim 18, wherein the receiver comprises a differential amplifier and, in the second mode, the third terminal is coupled to the differential amplifier to receive the second data stream.

20. The method of claim 15, wherein the first voltage level and the second voltage level are in a non-conducting voltage region of an electrostatic discharge protection circuit.

* * * * *